

Performance and Robustness Benchmarking Across Battery SOC Estimator Classes for Embedded Applications

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Abstract

Reliable deployment of neural battery-state estimators requires evidence beyond nominal prediction error. This study presents a causal, cell-disjoint benchmarking framework for the engineering validation of recurrent neural State-of-Charge (SOC) estimation against structurally distinct alternatives under identical data, disturbances, and embedded constraints. The comparison comprises fixed-capacity Coulomb counting as a direct-measurement model (DM), Coulomb counting with data-driven State-of-Health (SOH) adaptation as a hybrid direct-measurement model (HDM), a second-order resistor-capacitor observer as a hybrid equivalent-circuit model (HECM), and a gated recurrent neural estimator as the data-driven model (DD). Experiments use a public lithium iron phosphate ageing dataset spanning multiple load, thermal, and SOH conditions. The robustness benchmark separates nominal accuracy, resistance to error growth under corrupted or unavailable measurements, and recovery after initialization mismatch. The hardware benchmark verifies numerical equivalence and measures inference latency, flash occupancy, and peak runtime random-access memory on an STM32H753 microcontroller. Nominal cell-macro mean absolute errors are 0.0690, 0.0412, 0.0337, and 0.0258 for DM, HDM, HECM, and DD,

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respectively. DD achieves the lowest nominal error and leads several disturbance scenarios. This behavior is consistent with the benefit of learned multichannel temporal context, whereas HECM provides the strongest observed recovery profile. All four isolated SOC implementations reproduce their software references, fit the available memory, and meet the one-second sampling deadline. The findings demonstrate that recurrent neural SOC estimation can combine strong disturbance behavior with feasible embedded execution. The framework provides a reproducible engineering methodology for validating artificial-intelligence-based battery estimators against conventional baselines across estimation quality, robustness, recovery, and deployment cost.

Keywords: Battery management system, State of charge, Robustness benchmark, Fault tolerance, Equivalent circuit model, Recurrent neural network, Embedded implementation, Microcontroller

1 Nomenclature

2 *Acronyms*

3	BMS	Battery management system
4	SOC	State of charge
5	SOH	State of health
6	DM	Direct-measurement model
7	HDM	Hybrid direct-measurement model
8	HECM	Hybrid equivalent-circuit model
9	DD	Data-driven neural model
10	ECM	Equivalent circuit model
11	EKF	Extended Kalman filter
12	MAE	Mean absolute error
13	RMSE	Root-mean-square error
14	ADC	Analog-to-digital converter
15	CV	Constant-voltage phase

16	OCV	Open-circuit voltage
17	GRU	Gated recurrent unit
18	LSTM	Long short-term memory
19	LFP	Lithium iron phosphate
20	<i>Symbols</i>	
21	t	Time
22	$I(t)$	Current signal at time t
23	$U(t)$	Voltage signal at time t
24	$T(t)$	Temperature signal at time t
25	$\widehat{\text{SOC}}(t)$	Estimated state of charge at time t
26	$\widehat{\text{SOH}}(t)$	Estimated state of health at time t
27	$\text{SOC}_{\text{ref}}(t)$	Dataset SOC ground truth at time t
28	$\text{SOH}_{\text{ref}}(t)$	Reference SOH target at time t
29	g_I	Multiplicative current-gain error
30	ΔI	Additive current offset
31	ΔU	Additive voltage offset
32	ΔT	Additive temperature offset
33	σ_I	Standard deviation of injected current noise
34	σ_U	Standard deviation of injected voltage noise
35	σ_T	Standard deviation of injected temperature noise
36	Q_c	Online cumulative charge state used by the SOC models
37	C_{nom}	Nominal capacity
38	C_{eff}	Effective capacity used inside the estimator
39	U_{max}	Upper voltage threshold used for full-charge detection
40	U_{tol}	Voltage tolerance used in the full-charge condition
41	U_1, U_2	Polarization voltages of the first and second RC branch
42	ΔMAE	Error increase relative to baseline

43 1. Introduction

44 1.1. Motivation and practical relevance

45 Reliable battery state estimation is a core BMS function because SOC and
46 SOH determine safety margins, usable energy, operating limits, charge control, and degradation-aware system management in both mobile and stationary
47 lithium-ion battery applications [1, 2, 3]. Practical deployment quality is not
48 defined by clean-condition accuracy alone, because real BMS operation is exposed to sensor gain and offset errors, additive noise, initialization uncertainty
49 after restart, missing samples, irregular sampling, communication interruptions, and transient signal spikes caused by measurement-chain limitations or disturbances in the surrounding system [4, 5, 6]. The relevant engineering question
50 is therefore not only whether an estimator achieves low MAE under nominal measurements, but also whether it converges back to a reliable state after disturbed initialization and whether it remains stable under physically plausible
51 measurement corruption.

52 As summarized conceptually in Figure 1, the present benchmark treats
53 deployment-facing estimator quality as a three-dimensional problem consisting
54 of nominal accuracy, recovery after state inconsistency, and robustness against
55 disturbed measurements and signal-integrity faults, embedded within broader
56 constraints on memory and execution time. The central question is how four
57 structurally different SOC-estimation classes, represented by the implementations studied here, respond to common failure mechanisms and microcontroller
58 constraints, and whether their nominal ranking remains informative under those
59 conditions. This distinction is essential because one implementation can be accurate under clean measurements yet difficult to re-anchor, sensitive to a specific
60 input fault, or too costly under its trained inference semantics.

69 1.2. State of the art in battery state estimation

70 The literature commonly groups battery-state estimators into direct-measurement
71 or integration-based approaches, model-based approaches based on equivalent-circuit or electrochemical models, hybrid approaches that combine physical
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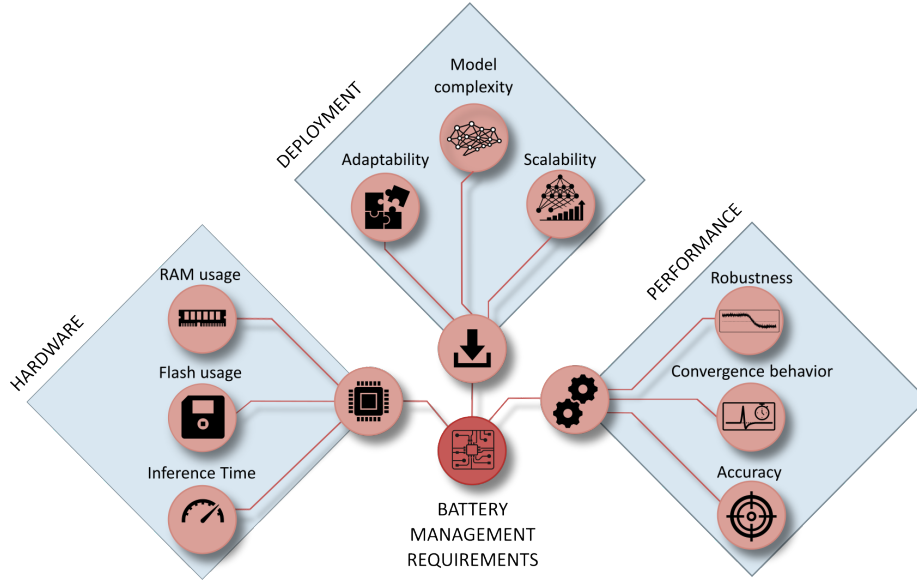


Figure 1: Conceptual overview of deployment-oriented requirements for embedded BMS state estimation. The broader requirement space includes implementation effort, hardware constraints, and estimator performance, whereas the present study focuses on the performance branch and evaluates baseline accuracy, convergence behavior, and robustness under measurement disturbances.

73 structure with learned components, and purely data-driven models based on
74 neural sequence learning [7, 1, 2, 3]. Direct-measurement approaches, typically
75 based on Coulomb counting, are attractive because they are simple, compu-
76 tationally inexpensive, and easy to deploy. However, they remain structurally
77 sensitive to current-gain error, missing data, and initialization errors because
78 they lack an internal correction mechanism [7, 4]. Model-based approaches,
79 especially ECM- and observer-based estimators, offer physical interpretability
80 and voltage-based correction capability, but their practical behavior depends
81 strongly on the quality of model parametrization, operating-point observability,
82 and measurement quality [8, 2, 4].

83 Hybrid estimators seek to combine physical state propagation with learned
84 adaptation of capacity, parameters, or auxiliary states, which has made joint
85 SOC/SOH estimation an active research direction [9, 10, 11]. This combination

can improve baseline calibration, while it does not necessarily eliminate structural sensitivity to measurement disturbances. Data-driven approaches based on recurrent or sequence models can exploit nonlinear cross-relations between current, voltage, temperature, and derived online features, and recent work has also demonstrated their embedded feasibility through TinyML-oriented implementations, pruning, and quantization [12, 13, 14, 15, 3].

1.3. Performance and embedded deployment gap

A large share of the SOC estimation literature still emphasizes clean-data accuracy, average error metrics, or computational efficiency, while comparatively fewer studies analyze how estimators behave under realistic measurement faults and timing corruption [1, 13, 14, 15]. Existing robustness studies often remain confined to a single model family, for example observer-based robustness against modeling and measurement errors [4] or robust estimation concepts for specific BMS architectures under noise and uncertainty [5], whereas cross-family comparisons under the same disturbance protocol and under comparable embedded constraints remain limited. Benchmarking studies in the broader BMS context underline the importance of reproducible scenario design and deployment-relevant validation, but they do not by themselves close the gap between clean-data estimator comparison and measurement-level performance benchmarking across fundamentally different estimator classes [6]. This gap is especially relevant in embedded BMS design, where estimator selection must jointly consider nominal accuracy, convergence, robustness under disturbed measurements, and computational constraints.

1.4. Contribution and study objectives

This work addresses the gap through three linked questions: whether clean-data accuracy predicts performance under measurement and signal-integrity disturbances, which structural mechanisms govern degradation and recovery, and how the resulting behavior trades against embedded execution cost. To answer them, four deployment-oriented representatives are executed under one

causal protocol on cell-disjoint LFP aging data covering multiple load and SOH states. The campaign combines physically interpretable measurement, timing, availability, quantization, and initialization disturbances with a common recovery criterion, cell-level uncertainty analysis, a paired SOH-input ablation, and STM32 measurements of the isolated SOC cores.

The four evaluated implementations provide distinct update and correction mechanisms whose behavior can be traced across the same disturbances, recovery criterion, and hardware measurements. The resulting comparison links estimator selection to the fault profile and resource constraints of the target BMS application.

2. Methodology

2.1. Estimator representatives and class boundaries

Figure 2 summarizes the causal benchmark chain, from measured signals and online feature reconstruction to disturbance injection and global and local evaluation. The four estimators were selected because their dominant SOC update and correction mechanisms differ. Table 1 relates these mechanisms to the broader design space and specifies the implementation evaluated for each class. Learned corrections, higher-order dynamics, richer sensing, and more complex sequence architectures extend these class envelopes while also changing deployment cost.

Table 1: Estimator-class envelope and implementation scope. Literature in the second column supports the defining mechanisms and broader capability ranges, while the last column specifies the representative evaluated in this study.

Class	Defining mechanism and broader capability range	Representative used in this study
Direct measurement	Causal integration of measured current using an assumed capacity. Extensions can include coulombic efficiency, capacity adaptation, OCV or end-point anchoring, plausibility logic, and reset thresholds [7, 13].	Fixed-capacity Coulomb counting with a sustained full-charge CV reset and no voltage-residual correction
Hybrid direct measurement	Integration core augmented by an estimated or learned state or parameter. Extensions can include learned capacity or efficiency, residual correction, multi-sensor fusion, and adaptive anchoring [10, 11].	Same Coulomb-counting and reset logic as DM, with effective capacity supplied by the shared causal LSTM-SOH estimate
ECM observer	Coulomb-counted SOC propagation combined with a voltage model and feedback correction. Extensions can include higher RC order, hysteresis, thermal states, operating-point maps, and different nonlinear observers [7, 4, 2].	Second-order RC model with EKF correction, charge/discharge SOC-SOH lookup surfaces, and shared causal SOH input

Table 1: Estimator-class envelope and implementation scope (continued).

Class	Defining mechanism and broader capability range	Representative used in this study
Data driven	Learned mapping from measured signals or their history to SOC. Implementations can use feed-forward, convolutional, recurrent, attention-based, physics-informed, raw-signal, or engineered-feature models [12, 15, 13].	Structurally pruned rolling-window GRU using voltage, current, temperature, SOH, Q_c , derivatives, and Δt

135 All SOH-dependent representatives, namely HDM, HECM, and DD, receive
 136 the same causal LSTM-SOH trace so that differences arise from how the SOC
 137 branch consumes that context. DM is the low-complexity integration reference.
 138 HDM isolates the effect of learned capacity adaptation without adding volt-
 139 age feedback. HECM adds model-based voltage-residual correction to current-
 140 driven state propagation. DD represents compact temporal learning. The GRU
 141 was selected because SOC and disturbance recovery depend on recent signal his-
 142 tory while its gated single-state structure is lighter than a comparable LSTM.
 143 Because DD also consumes the engineered Q_c feature, it is a data-driven se-
 144 quence estimator with physics-informed inputs rather than a raw-signal-only
 145 neural model.

146 2.2. SOC estimation methods

147 Direct-measurement model (DM)

148 The direct-measurement SOC formulation used in DM, HDM, and as an
 149 explicit auxiliary feature in DD follows the same online cumulative-charge state,
 150 denoted here uniformly as Q_c . It is updated causally from the measured current
 151 stream as

$$Q_c(k) = Q_c(k-1) + \frac{I_k \Delta t_k}{3600 \text{ s h}^{-1}}, \quad (1)$$

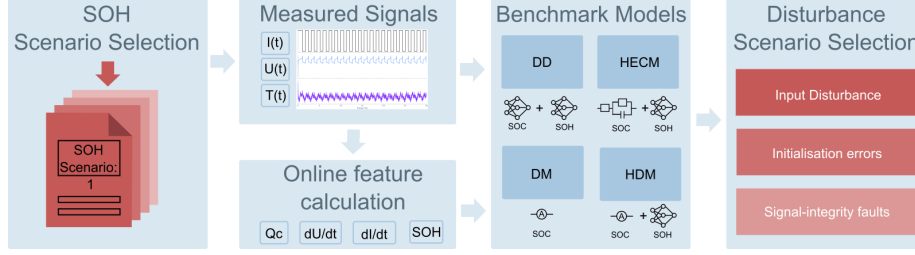


Figure 2: Methodology overview for the causal benchmark chain. Measured battery signals are converted into online auxiliary features, processed by representative estimator implementations, exposed to measurement and initialization interventions, and finally evaluated through global and local performance metrics.

where I_k is given in ampere and Δt_k in seconds, so that the factor 3600 s h^{-1} converts the increment to ampere-hours. The sign convention is fixed by the benchmark implementation and the state is reset to $Q_c = 0$ after a sustained constant-voltage full-charge event. In the DM, this same charge state is mapped directly to SOC through the fixed nominal capacity

$$\widehat{\text{SOC}}_{\text{DM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{nom}}}, 0, 1\right). \quad (2)$$

Hybrid direct-measurement model (HDM)

The HDM preserves exactly the same Coulomb-counting core and differs only in the capacity term. The SOH estimate rescales the usable capacity as

$$C_{\text{eff}}(k) = C_{\text{nom}} \cdot \widehat{\text{SOH}}(k), \quad (3)$$

which yields

$$\widehat{\text{SOC}}_{\text{HDM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{eff}}(k)}, 0, 1\right). \quad (4)$$

This formulation makes the relation between DM and HDM explicit: both use the same online charge state Q_c , while only the denominator changes through online SOH support. In both direct-measurement variants, a sustained near-maximum voltage condition is interpreted as a constant-voltage full-charge event. If the voltage remains above $U_{\text{max}} - U_{\text{tol}}$, where U_{tol} denotes the admissible voltage margin below the upper threshold U_{max} , for the configured CV duration,

the internal charge state is reset so that $Q_c = 0$ and therefore $\widehat{\text{SOC}} = 1$. This reflects the practical assumption that a completed CV phase provides a physically anchored full-charge reference.

Hybrid equivalent-circuit model (HECM)

The hybrid ECM model is implemented as a second-order RC model with the state vector

$$\mathbf{x}_k = \begin{bmatrix} \widehat{\text{SOC}}_k & U_{1,k} & U_{2,k} \end{bmatrix}^\top, \quad (5)$$

where U_1 and U_2 denote the polarization voltages of the two RC branches. The prediction step propagates the state according to

$$\mathbf{x}_k^- = \mathbf{A}_k \mathbf{x}_{k-1} + \mathbf{b}_k I_{k-1}, \quad (6)$$

with

$$\mathbf{A}_k = \text{diag}\left(1, e^{-\Delta t_k/\tau_1}, e^{-\Delta t_k/\tau_2}\right), \quad \mathbf{b}_k = \begin{bmatrix} \eta \Delta t_k / C_b \\ R_1(1 - e^{-\Delta t_k/\tau_1}) \\ R_2(1 - e^{-\Delta t_k/\tau_2}) \end{bmatrix}, \quad (7)$$

where $C_b = C_0 \widehat{\text{SOH}}$ is the SOH-scaled available capacity. The corrected terminal-voltage estimate is written as

$$\hat{U}_k = \text{OCV}(\widehat{\text{SOC}}_k, \widehat{\text{SOH}}_k, I_k) + U_{1,k} + U_{2,k} + R_i I_k, \quad (8)$$

and the Kalman update uses the residual between $\hat{U}_k$ and the measured terminal voltage to correct the internal state. In the present implementation, the state propagation is driven by the previous current sample I_{k-1} , whereas the measurement equation uses the current sample I_k available at the correction step. The ECM parameters are not constant. The implementation interpolates R_i , R_1 , R_2 , τ_1 , τ_2 , OCV, and dOCV/dSOC from a preidentified lookup table over SOC and SOH, with separate charge and discharge parameter surfaces. The HECM is implemented as a two-RC observer with SOH-dependent parameter scheduling.

2.2. Data-driven SOC model (DD)

The data-driven SOC model is a GRU-based recurrent estimator with the feature vector $\{U \text{ [V]}, I \text{ [A]}, T \text{ [}^\circ\text{C]}, \text{SOH}, Q_c, dU/dt, dI/dt, \Delta t\}$. The benchmarked, structurally pruned and fine-tuned model consists of a single-layer GRU with hidden size 67 followed by an MLP head with one hidden layer of size 96 and a sigmoid output layer. During optimization, the recurrent core is exposed to causal sequence segments of length $L_{\text{SOC}} = 2024$ samples. Within this feature set, Q_c is exactly the same online cumulative-charge state defined in (1). It is included explicitly so that the neural estimator receives the same charge-throughput information that also drives DM and HDM. The derivative channels are computed online as the finite differences $dI/dt(k) = (I_k - I_{k-1})/\Delta t_k$ and $dU/dt(k) = (U_k - U_{k-1})/\Delta t_k$. They provide local transient information that is not captured by the absolute channels alone. The additional feature Δt_k allows the network to distinguish amplitude changes from sampling-interval changes under irregular timing. During the primary benchmark, each output is calculated from the latest causal 2024-sample window and recurrent state is reset between windows, matching training and validation. This configuration uses measured and causally reconstructed inputs up to the current sample. The hardware analysis additionally evaluates continuous hidden-state forwarding as a deployment variant and reports its accuracy and resource demand separately.

2.3. SOH estimation method

The shared SOH estimator is an LSTM-based sequence-to-sequence model that operates on hourly aggregated online features derived from $\{U \text{ [V]}, I \text{ [A]}, T \text{ [}^\circ\text{C]}, \text{EFC}, Q_c\}$, where the aggregation uses the statistics mean, standard deviation, minimum, and maximum for each feature [16, 17, 18, 19]. This transforms the five base channels into a 20-dimensional hourly feature vector. Architecturally, the SOH network first projects the aggregated input through a two-layer feature embedding block with size 128, layer normalization, GELU activations, and dropout, then processes the sequence with a two-layer unidirectional LSTM of hidden size 112, and finally maps the recurrent output through

three residual MLP blocks and a prediction head with hidden size 160 to obtain the SOH estimate for each hourly step.

During optimization, the SOH estimator is exposed to causal hourly sequence segments, and benchmark execution processes one hourly aggregate at a time with causal hidden- and cell-state forwarding. This allows the estimator to accumulate degradation context without access to future information. The first hourly output is anchored with the configured initial SOH value before subsequent hourly predictions are taken directly from the recurrent model output. The SOH prediction is updated at hourly cadence and held piecewise constant between update points. This online update structure matches the gradually evolving capacity fade in the dataset and supplies the latest SOH context to the faster SOC estimators. In the HDM, the predicted SOH enters the SOC computation through the effective capacity relation $C_{\text{eff}} = C_{\text{nom}} \cdot \widehat{\text{SOH}}$, so the method remains structurally close to direct measurement while replacing the fixed-capacity assumption. In the HECM, the predicted SOH affects the capacity scaling and the lookup-based ECM parametrization, while in the DD model it is provided as an explicit SOC input feature that contextualizes the current measurement trajectory.

A paired diagnostic comparison is used to distinguish errors produced within each SOC branch from errors inherited through the estimated SOH input. For selected scenarios, the causal LSTM-SOH trace is replaced by the offline dataset ground truth while the SOC implementation, disturbed measurements, evaluation window, and error calculation remain unchanged. This substitution is not a deployable estimator configuration. It isolates how strongly each SOC branch depends on the accuracy and characteristics of the supplied health information.

2.4. Model and target-hardware preparation

The software benchmark uses trained floating-point model parameters and input scalers fixed before evaluation on the independent holdout test set. Neural training, hyperparameter selection, scaler fitting, pruning, and model selection use the declared training and validation sets. The HPPC-derived HECM lookup

surfaces were identified from the same model-development sets and fixed without retuning before benchmark evaluation. Figure A.25 and Table A.4 document the complete cell-disjoint split. The compact SOC-GRU has hidden size 67 and the shared SOH-LSTM has hidden size 112. The continuous high-load test trajectory in Figure 6 is analyzed separately as a mechanism-focused lifecycle diagnostic.

The target-hardware experiment evaluates the isolated SOC cores on a NUCLEO-H753ZI board with a 480 MHz STM32H753 Cortex-M7. To separate SOC implementation cost from the shared SOH service, the board receives the precomputed causal SOH trace used by the software reference. The SOH-LSTM itself is not executed on the microcontroller. Numerical equivalence and inference latency are measured directly on the board. Execution time is recorded for every valid inference with the processor-integrated hardware cycle counter, and each cell sequence is replayed three times. Flash occupancy is calculated from the compiled firmware regions that must be stored in the microcontroller’s non-volatile program memory.

Peak runtime RAM combines statically allocated variable storage, maximum dynamic-memory allocation, and maximum temporary call-stack demand. The static contribution is obtained from the compiled firmware, while dynamic allocation is monitored during execution. Stack demand is measured with a high-water-mark pattern during representative on-device replays that exercise the complete estimator path. For DD, this includes valid inference after its rolling input window has been filled. An unused compilation-time memory reserve is excluded from the reported demand. Predictions are compared with both the software implementation and dataset SOC. Figure 19 summarizes the resulting memory requirements of the isolated SOC implementations with SOH supplied as an external causal trace.

2.5. Robustness benchmark design

The primary cross-model benchmark manipulates current, voltage, temperature, sample availability, and sampling time, all of which are measured or recon-

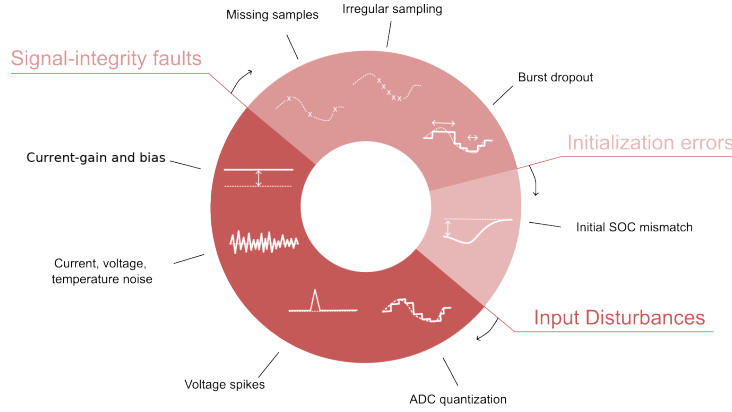


Figure 3: Taxonomy of the disturbance scenarios used in the performance benchmark, grouped into input disturbances, initialization errors, and signal-integrity faults.

277 structured online by a BMS. A separate initialization branch acts on deployment-
 278 accessible estimator states. Parameter mismatch is excluded from this common
 279 ranking and examined through the supplementary HECM analysis defined in
 280 Section 2.7. Figure 3 groups the primary scenarios into input disturbances,
 281 initialization errors, and signal-integrity faults, while Table 2 defines their im-
 282 plementation [6].

283 The disturbance magnitudes follow three practical levels of justification:
 284 deployment-level sensing limits, adverse measurement corruption, and signal-
 285 integrity stress conditions [20, 21, 22, 23, 24, 25]. Measurement disturbances
 286 are separated by mechanism. Random sensor noise represents zero-mean fluc-
 287 tuations in current, voltage, and temperature. Systematic sensor errors com-
 288 prise multiplicative current-gain error and additive offsets in current, voltage,
 289 and temperature. Voltage spikes and ADC quantization complete the input-
 290 disturbance branch. The remaining scenarios address estimator initialization,
 291 sample availability, and sampling-time integrity. Together they form a con-
 292 trolled and reproducible benchmark matrix for exposing estimator-specific fail-
 293 ure mechanisms under a common protocol.

Table 2: Disturbance configurations used in the benchmark and their practical rationale. Exact numeric values are fixed benchmark settings. The cited sources motivate disturbance type and magnitude range.

Scenario	Benchmark setup	Practical origin / rationale
ADC quantization	$\Delta I = 0.01$ A, $\Delta U = 0.005$ V, and $\Delta T = 0.5$ °C	ADC steps, sensor scaling, rounding, and coarse deployment-level discretization [20, 21, 26]
Current gain error	$I_{\text{meas}} = I(1 + g_I)$ with $g_I = \pm 0.5\%$, $\pm 1.5\%$, and $\pm 3.0\%$. Both signs are evaluated at each magnitude	Sensor sensitivity error, scale-factor miscalibration, drift, and upper bound as stress case [21, 22, 23]
Current noise	Gaussian with $\sigma_I = 0.02$ A and 0.10 A	Sensing-chain noise, EMI pickup, and low and high sensor-noise levels [4, 5]
Voltage noise	Gaussian with $\sigma_U = 0.01$ V	Wiring noise, front-end noise, poor filtering, and EMI [4, 20, 24]
Temperature noise	Gaussian with $\sigma_T = 1.0$ °C	Sensor tolerance, conversion noise, thermal gradients, and sensor-to-cell mismatch [26, 11]
Current offset	Additive $I_{\text{meas}} = I + \Delta I$ with $\Delta I = \pm 0.05$ A. Both signs are evaluated	Zero-current calibration residual and systematic current-sensor offset [21, 22, 23]
Voltage offset	Additive $U_{\text{meas}} = U + 0.02$ V	Front-end offset, calibration residual, wiring contribution, and adverse sensing stress
Temperature offset	Additive $T_{\text{meas}} = T + 3$ °C	Sensor placement error, calibration residual, and cell-to-sensor thermal mismatch
Initial SOC error	Initial mismatch of -0.10 SOC, with DD perturbed through online Q_c	Restart, storage, missing recent history, and weak LFP OCV anchoring [4, 8]

Table 2: Disturbance configurations used in the benchmark and practical rationale (continued).

Scenario	Benchmark setup	Practical origin / rationale
Missing samples	Every 50th sample frozen and separate random 2% loss	Packet loss, blocked updates, logger dropouts, and periodic and random data freeze [5, 25]
Irregular sampling	Uniform jitter: ± 0.1 s, ± 0.5 s, and ± 0.9 s around nominal 1 s grid	Scheduling jitter, bus latency, and asynchronous acquisition [6, 5, 25]
Burst dropout	One frozen 3600 s interval selected by maximum absolute net charge, with at least 12 h before and 24 h after the interval	Communication interruption, frozen task, stuck data stream, and measurement-only severe case [6, 5]
Voltage spikes	Single-sample spikes of ± 0.20 V every 1000 samples (1000 s)	EMI, switching disturbance, and corrupted voltage samples [24]

2.6. Metrics and evaluation criteria

For deployment-oriented interpretation, the benchmark results are synthesized along three performance dimensions: nominal accuracy under baseline conditions, recovery after initialization mismatch, and robustness under sustained or transient measurement corruption and signal unavailability. Disturbance-related global performance is quantified through MAE, RMSE, bias, maximum error, and the 95th percentile absolute error, because these metrics provide a compact view of average performance, systematic deviation, and tail behavior across the selected evaluation windows. Local disturbance performance is analyzed through scenario-specific metrics such as jump counts, output variance, absolute-error variance, and excess rolling MAE, because several disturbances do not primarily manifest as window-wide MAE collapse but as short-term output fluctuations, delayed drift, or slow re-synchronization. Whenever a disturbance

mask exists, the analysis further separates pre-disturbance, disturbed, and post-disturbance behavior in order to quantify disturbance-window error growth and residual post-disturbance error.

ADC quantization, current-gain error, additive sensor offsets, voltage and temperature noise, missing samples, and irregular sampling are interpreted primarily through global metrics. Burst dropout, voltage spikes, current-noise detail, and initialization mismatch additionally use dedicated local evidence because their most relevant effects are episodic. The canonical initialization-recovery analysis compares each perturbed estimator trajectory with its cell-, window-, model-, SOH-, and seed-matched correctly initialized trajectory. The resulting paired trajectory difference is the recovery signal, while absolute error against dataset SOC remains the accuracy measure. A first qualifying entry occurs when the paired difference enters a 0.02 SOC band for at least 300 s. Persistent recovery is the first subsequent point after which the difference remains within this band for the rest of the 24-h observation horizon. Runs without this endpoint are right-censored. First-entry time, later departure from the band, censored and relapse fractions, and excess-error area under the curve provide complementary recovery diagnostics. The protocol maps a -0.10 SOC-equivalent mismatch into the coulomb-counting SOC initialization for DM and HDM, the EKF SOC state for HECM, and the capacity-equivalent Q_c offset for DD. Because DD first requires a causal input context, recovery is evaluated for every estimator from the first time at which all four provide a valid output. Recovery times remain referenced to the original intervention. Endpoints already satisfied at the common evaluation start are treated as left-censored and recorded as conservative upper bounds. The comparison therefore measures observable re-anchoring after deployment-accessible, estimator-specific initialization interventions.

Each holdout cell is treated as one independent statistical unit. Protocol-defined SOH windows and random seeds are first summarized within each cell, after which the six cells receive equal macro weight. Hierarchical bootstrap resampling with seeds nested inside cells and 10,000 repetitions provides 95%

confidence intervals. Scenario effects are evaluated as paired, cell-matched differences from baseline, and estimator pairs are compared using exact two-sided sign-flip tests, raw mean and median differences, paired standardized effect sizes, and Holm-adjusted p -values. Holm correction is applied within each reported scenario family. With six independent cells, effect magnitudes and confidence intervals provide the primary evidence and p -values support their interpretation. The full machine-readable tests accompany the result tables, while the baseline pair tests are reproduced in the Appendix.

To ensure like-for-like comparisons, all global accuracy, robustness, stratified, confidence-interval, and paired-test calculations begin when the rolling DD first provides a valid estimate. The resulting common evaluation mask gives every estimator the same source interval and the same number of scored observations.

The decision synthesis provides a relative summary of the four implementations. Accuracy combines baseline MAE, RMSE, and 95th-percentile error. Robustness assigns equal weight to eight evaluated disturbance families: random sensor noise, current-gain error, additive sensor offsets, quantization, missing samples, timing jitter, burst dropout, and voltage spikes. The current-gain and current-offset components use the adverse direction from their matched signed sweeps. Non-positive Δ MAE values contribute zero disturbance penalty. Levels within a family are averaged before the family scores are combined, giving every disturbance family equal influence. Recovery combines observed persistent recovery-or-censor time, excess-error area, and persistent censored fraction from the canonical paired initialization analysis. Relapse after a first qualifying entry remains a separate diagnostic because it is defined only for runs that enter the recovery band. Equal-scenario and high-severity weighting variants quantify the sensitivity of the summary to alternative deployment priorities.

2.7. HECM lookup-table sensitivity analysis

The HECM is the only tested representative whose SOC calculation depends directly on identified resistance, time-constant, and OCV lookup surfaces. Consequently, an observed HECM disturbance penalty could originate from the

manipulated measurement, from local lookup-table calibration uncertainty, or from an interaction between both. The sensitivity analysis tests whether the HECM robustness profile remains stable when these lookup surfaces are varied and whether the larger disturbance penalties can still be attributed primarily to the disturbed measurements. This separates measurement robustness from parameter-calibration sensitivity and is necessary for interpreting the HECM benchmark results.

The nominal lookup surfaces are identified from the training and validation data and remain fixed during the primary cross-model benchmark. For the sensitivity test, the resistance surfaces and the time-constant surfaces are each scaled independently by $\pm 10\%$, while the OCV surfaces are shifted by ± 10 mV. Each one-at-a-time lookup perturbation is crossed with the complete set of measurement and signal-integrity subcases, including the signed current-offset test, and with the paired initialization-recovery experiment. Because these lookup perturbations apply only to HECM, this analysis is evaluated separately and does not enter the cross-model robustness score.

For every disturbance, the lookup interaction is calculated as the change in scenario ΔMAE caused by the perturbed lookup relative to the corresponding result with the nominal lookup. Results are first aggregated within cells and then summarized with the same equal-weight cell bootstrap as the primary benchmark. This design separates sensitivity to local HECM calibration changes from sensitivity to the disturbed measurement itself. The complete numerical protocol and the supporting figure are provided in Appendix Appendix B.

3. Experimental Setup

3.1. Dataset and data acquisition

The benchmark uses the MGFarm LFP 18650 dataset because its controlled design-of-experiments aging study provides long 1-Hz full-cell trajectories, repeated capacity checkups, multiple operating conditions, and enough independent cells for a cell-disjoint comparison [27, 28, 29]. The fixed split comprises

397 a six-cell training set, a three-cell validation set, and a six-cell independent
 398 holdout test set. Figure A.25 shows the complete SOH trajectories together
 399 with their charge-rate, discharge-rate, and depth-of-discharge aging conditions.
 400 Table A.4 provides the exact cell-to-set assignment and reports the measured
 401 duration, SOH, temperature, and load coverage. Figure A.21 summarizes the
 402 diversity within the holdout test set. Together, these materials document the
 403 cell-disjoint development boundary and the operating diversity of the evalua-
 404 tion data. LFP’s flat mid-SOC OCV and chemistry-specific aging behavior are
 405 integral to the tested problem.

406 The holdout test set is assigned to descriptive load groups using the measured
 407 95th-percentile absolute C-rate. It contains two low-load cells, three middle-
 408 load cells, and one high-load cell. The single high-load trajectory is exploratory,
 409 whereas the low- and middle-load groups support multi-cell summaries. The ex-
 410 act assignment and the corresponding load coverage are shown in Figure A.21
 411 and Table A.4. These cell-level load groups describe the operating coverage of
 412 the holdout set and are distinct from the multivariate selection of representa-
 413 tive evaluation windows. The latter uses measured SOH, temperature, C-rate,
 414 charge-throughput, and SOC-coverage descriptors as detailed in Section 3.3.
 415 This separation preserves variation across cells and aging states without treat-
 416 ing every operating-state bin as an independent statistical sample.

417 3.2. Dataset ground truth and preprocessing

418 The reference preprocessing first computes the signed transferred charge as
 419 $Q_k = I_k \Delta t_k / 3600$ and the absolute transferred charge as $|Q_k|$, from which the
 420 capacity during dedicated capacity-test intervals is obtained as $C_{\text{ref}} = \sum_k |Q_k|$.
 421 The reference SOH target is then defined offline as $\text{SOH}_{\text{ref}} = C_{\text{ref}} / C_{\text{ref},0}$, where
 422 $C_{\text{ref},0}$ denotes the first measured reference capacity of the trajectory, and the
 423 resulting capacity and SOH columns are interpolated between the dedicated
 424 capacity-test anchors. The dataset SOC ground truth used in the benchmark
 425 is reconstructed offline as $\text{SOC}_{\text{ref}} = 1 + Q_c / C_{\text{ref},0}$, where the cumulative charge
 426 state Q_c is propagated from the signed charge and reset to 0 at the upper

427 voltage anchor and to $-C_{\text{ref},0}$ at the lower voltage anchor during preprocessing.
 428 This reconstructed SOC serves as the common dataset ground truth for every
 429 reported SOC error. It is an offline evaluation quantity because its dataset-side
 430 processing and anchor logic are not fully available to an online estimator during
 431 deployment. All nominal accuracy values therefore refer to this consistently
 432 applied dataset ground truth. Paired ΔMAE compares each disturbed run with
 433 its matched baseline under the same ground truth.

434 3.3. Test trajectory and benchmark scenarios

435 All estimators are evaluated on every available cell/SOH combination in the
 436 holdout test set under the same disturbance protocol. One representative 24-
 437 h window is frozen for each available fresh, mid-life, and aged state, yielding
 438 16 windows. One test trajectory remains entirely within the fresh SOH range
 439 and therefore contributes no mid-life or aged window, as shown in Figure A.21.
 440 Every candidate window begins at the same measured full-charge condition
 441 ($\text{SOC} \geq 0.98$, $U \geq 3.58\text{ V}$). Multiple eligible windows can nevertheless exist for
 442 one cell and SOH state, and their temperature, load, charge throughput, and
 443 SOC coverage can differ substantially. Each candidate window w is therefore
 444 represented by the measured operating-condition vector

$$\mathbf{z}_w = \left[\widetilde{\text{SOH}}_w \quad \bar{T}_w \quad P_{95}(T)_w \quad P_{95}(|C_{\text{rate}}|)_w \quad Q_{\text{through},w} \quad f_{\text{SOC} < 0.2,w} \right]^\top, \quad (9)$$

445 where the entries denote median SOH, mean temperature, 95th-percentile tem-
 446 perature, 95th-percentile absolute C-rate, absolute charge throughput, and the
 447 fraction of samples below 20% SOC. For every descriptor j , the candidate values
 448 are centered by their median m_j and scaled by their median absolute deviation
 449 s_j . The standard deviation is used when the median absolute deviation is zero,
 450 and an invariant descriptor receives unit scale. The selected observed window
 451 is

$$w^* = \arg \min_w \sqrt{\sum_j \left(\frac{z_{w,j} - m_j}{s_j} \right)^2}. \quad (10)$$

452 This measured-feature medoid criterion selects the actual window closest to the
 453 typical multivariate operating profile within each cell and SOH state. It avoids

arbitrary or estimator-favorable window selection because neither model predictions nor errors enter the criterion. Applying the same procedure separately to every holdout cell and available aging state retains diversity across cells and SOH conditions while preventing unusual operating periods from dominating the comparison.

The one-hour dropout uses 48 h from the same full-charge anchor to retain approximately 24 h of post-gap observation. The shared SOH LSTM receives 192 h of undisturbed causal context before every scored window, matching its trained sequence length. Figure A.23 summarizes the selection and evaluation protocol.

The benchmark contains 20 scenario definitions: the nominal baseline, two current-noise levels, voltage and temperature noise, three signed current-gain magnitudes, one signed additive current-offset level, voltage and temperature offsets, ADC quantization, initial SOC mismatch, periodic and random sample loss, three timing-jitter levels, a one-hour burst dropout, and randomized-sign voltage spikes. Each current-gain magnitude and the 50 mA current-offset case contain matched positive and negative sign sublevels. Adverse-direction summaries select the larger paired ΔMAE of the two signs within each cell before equal-weight aggregation. Gaussian sensor-noise cases use 10 deterministic seeds. Random sample loss, jitter, and spike cases use 5 seeds. Deterministic cases run once. The complete traceable cross-model build comprises 7040 model-window evaluations. Figure A.22 maps the scenario definitions to the manipulated measurement or state and marks the selected paired reference-SOH ablations. Input disturbances act directly on measured channels, initialization interventions act on each estimator’s deployment-accessible state, and signal-integrity scenarios alter sample availability or timing while preserving causal execution.

3.4. Implementation details

Every estimator is executed causally, and all auxiliary quantities required by the models are rebuilt directly from disturbed inputs. This applies to Q_c ,

484 EFC, dU/dt , dI/dt , and hourly SOH aggregates, ensuring that estimator inputs
 485 originate from the causal measurement path. The SOH LSTM forwards hidden
 486 and cell states across hourly updates after its 192-h initialization context. The
 487 primary SOC-GRU evaluation preserves its trained sequence-to-one semantics:
 488 each prediction uses the latest 2024-sample rolling window with recurrent state
 489 reset between windows. The continuous-stateful stream forms a separate de-
 490 ployment configuration whose full-trajectory accuracy and resource demand are
 491 reported alongside the rolling implementation. In freeze and dropout scenar-
 492 ios, disturbed or frozen measurements propagate consistently into derived online
 493 features. The dropout start is selected from measured current as the eligible one-
 494 hour interval with the largest absolute integrated net charge. Selection enforces
 495 at least 12 h of pre-gap observations and 24 h of post-gap observations and re-
 496 mains independent of dataset SOC and estimator output. Direct-measurement
 497 estimators use the same CV-triggered reset that defines the full-charge start,
 498 whereas the DD initial-SOC test uses an equivalent offset of online Q_c because
 499 the network exposes no explicit SOC state. The environment enforces common
 500 nominal-capacity metadata, disturbed streams, targets, masks, and metrics.
 501 Model-specific capacity handling remains part of the estimator definitions.

502 4. Results

503 4.1. Baseline performance

504 Figure 4 summarizes nominal accuracy against the dataset ground truth
 505 across the 16 predefined windows on the common evaluation interval. Equal
 506 weighting within the six-cell holdout test set gives the lowest baseline MAE to
 507 DD at 0.0258 (95% CI 0.0189–0.0325), followed by HECM at 0.0337 (0.0246–
 508 0.0434), HDM at 0.0412 (0.0301–0.0515), and DM at 0.0690 (0.0499–0.0823).
 509 The RMSE ranking is identical, with values of 0.0300, 0.0388, 0.0442, and
 510 0.0740, respectively. The ordering is consistent with the correction informa-
 511 tion available to the implementations. DM propagates charge with fixed ca-
 512 pacity and has no continuous feedback correction. HDM reduces aging-related

capacity mismatch through its SOH-dependent effective capacity. HECM additionally uses a voltage residual to correct its model-based state, while DD maps the complete multichannel history to SOC and can represent nonlinear temporal relations that are absent from the other implementations. These mechanisms explain the observed ordering within the tested data without implying that architecture alone determines accuracy in every operating domain. Results at the cell and SOH-window level reveal substantial variation across the holdout trajectories, particularly for DM, and place the cell-macro means in the context of heterogeneous operating and aging conditions. The agreement between MAE and RMSE establishes a consistent observed cell-macro ranking within the tested data. All six pairwise mean differences follow the same ranking direction, and five of the six paired 95% confidence intervals exclude zero. The effect magnitudes and uncertainty intervals therefore provide quantitative evidence of practical separation between the tested implementations. Exact sign-flip tests add a complementary check of directional consistency at cell level. With six independent cells, their Holm-adjusted p -values remain above 0.05 and are interpreted together with the effect magnitudes and confidence intervals rather than in isolation. Taken together, these statistics support a clear ranking for the tested implementations and operating envelope. The following scenarios complement this nominal comparison by revealing distinct strengths and weaknesses in robustness, recovery, and embedded deployment. All reported accuracy values refer to the consistently reconstructed dataset ground truth.

4.2. *Current-gain sensitivity*

Current-gain error is a systematic sensor error that persists over time and scales with the instantaneous current. It therefore differs both from additive offsets, which shift the measurement zero independently of signal magnitude, and from zero-mean random sensor noise. Figure 5 summarizes the matched signed current-gain sweep at $\pm 0.5\%$, $\pm 1.5\%$, and $\pm 3.0\%$. For each magnitude and holdout cell, the adverse-direction summary uses the larger MAE change from the two signs because one direction can partially compensate a pre-existing baseline

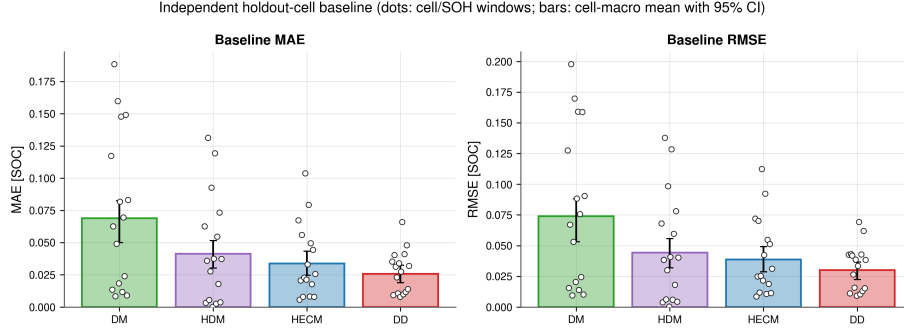


Figure 4: Baseline accuracy against the dataset SOC ground truth across the independent holdout test set. Dots show cell/SOH-window values. Bars and error bars show the equal-weight cell-macro mean and hierarchical 95% confidence interval. DD has the lowest macro MAE and RMSE, while DM has the largest error and between-cell spread.

error. The resulting degradation increases monotonically with gain-error magnitude for every estimator. DM and HDM show the strongest response, HECM is less sensitive, and DD is least sensitive. At 3%, the cell-macro MAE increases by 0.0108 for DM, 0.0101 for HDM, 0.0060 for HECM, and 0.0038 for DD. DM and HDM integrate the scaled current directly, so the error accumulates with transferred charge. HECM is exposed through the same current-driven state propagation, but its voltage-residual correction can counter part of the resulting state deviation. DD receives current together with voltage, temperature, cumulative charge, derivatives, and temporal context, which is consistent with its smaller observed dependence on one systematically scaled channel. The complete signed sweep remains available for interpreting direction-specific behavior.

The continuous high-load test-set analysis in Figure 6 complements the window-based benchmark and separates three mechanisms that independent 24-h windows cannot resolve. The gain-error contribution varies over life because its signed Coulomb-counting error interacts with charge direction and operating state. Between full charges, the penalty often grows within an interval, but full-charge anchoring can partially reduce or restructure it. The process is therefore neither an unrestricted monotonic drift nor a complete reset. Across

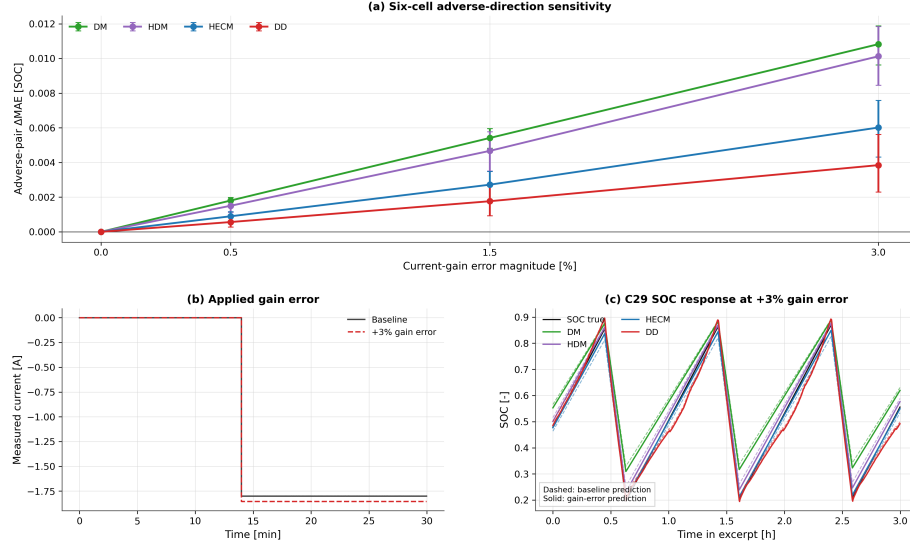


Figure 5: Current-gain sensitivity. (a) Six-cell adverse-direction ΔMAE for each matched $\pm g_I$ pair with hierarchical 95% confidence intervals. (b) Representative current before and after the +3% gain error. (c) SOC trajectories from the representative high-load test cell over three charge–discharge intervals. The test-set load assignment is documented in Figure A.21. The adverse-pair summary reports the larger penalty from each matched sign pair.

561 this single full-life replay, the adverse 3% gain error increases MAE by 0.0102 for
562 DM, 0.0072 for HDM, 0.0037 for HECM, and 0.0048 for DD. HECM therefore
563 has the smallest full-life gain-error contribution in this trajectory, while DM and
564 HDM remain most exposed. This mechanism-focused diagnostic is not a six-cell
565 lifecycle ranking.

566 4.3. Additive sensor offsets

567 The signed current-offset experiment shifts the measured current by ± 0.05 A
568 and selects the larger paired ΔMAE within each cell before equal-weight aggrega-
569 tion. DD has the smallest adverse cell-macro penalty at 0.0401 (95% CI
570 0.0226–0.0556), followed by HECM at 0.1763 (0.1329–0.2142), DM at 0.2467
571 (0.2253–0.2693), and HDM at 0.2727 (0.2546–0.2911). The adverse sign dif-
572 fers across models and cells, which confirms that a single favorable sign would
573 not represent offset sensitivity reliably. The large response of the integration-

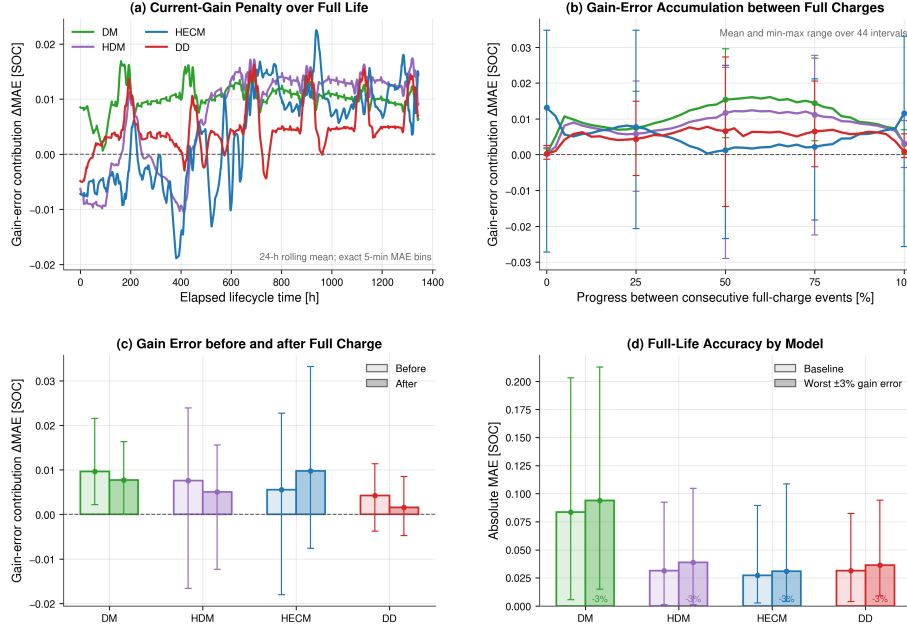


Figure 6: Current-gain-error diagnostic over the continuous high-load holdout trajectory. (a) Rolling 24-h gain-error contribution over elapsed life. (b) Mean and range over normalized intervals between consecutive full-charge events. (c) Penalty immediately before and after full-charge events. (d) Full-life baseline and adverse-gain MAE. Full-charge events reduce or restructure accumulated error but do not erase the lifetime penalty uniformly.

574 based implementations follows from accumulating a persistent zero-current error
 575 throughout the evaluation window, whereas the recurrent DD implementation
 576 limits but does not eliminate this systematic shift. Among the evaluated mea-
 577 surement disturbances, additive current offset produces the largest global cell-
 578 macro penalty for all four representatives. The complete comparison with the
 579 remaining disturbance scenarios is shown in Figure 15.

580 The fixed positive voltage and temperature offsets produce a different model
 581 pattern. A $+3^{\circ}\text{C}$ temperature offset increases DD MAE by 0.0074 (95% CI
 582 0.0027–0.0125), while the corresponding changes for DM, HDM, and HECM
 583 are 0.0000, 0.0002, and 0.0002. A $+0.02\text{ V}$ voltage offset increases DD by 0.0038

584 and HDM by 0.0005, leaves DM effectively unchanged, and changes HECM by
 585 -0.0040 in these finite windows. This pattern follows the model-specific use of
 586 the shifted channels. DM does not use temperature and uses voltage primarily
 587 for physical anchoring rather than continuous propagation. HDM receives both
 588 channels indirectly through the slowly updated SOH service. HECM responds
 589 to voltage through its observer residual, where the fixed offset can partly com-
 590 pensate an existing trajectory error over a finite window. DD uses absolute
 591 voltage and temperature as direct learned inputs and therefore transmits both
 592 systematic shifts into the SOC prediction. The signed HECM result must con-
 593 sequently be interpreted as compensation relative to its baseline rather than an
 594 improvement in the physical model. The additive current, voltage, and temper-
 595 ature offsets form one offset family in the robustness synthesis, with the signed
 596 current case represented by its adverse paired result.

597 *4.4. Noise sensitivity*

598 In contrast to the persistent gain and offset cases, the sensor-noise experi-
 599 ments represent random zero-mean fluctuations. They produce smaller global
 600 penalties than the systematic current errors, with a response that depends on
 601 the disturbed channel and estimator structure. At $\sigma_I = 0.10$ A, HECM has
 602 the largest mean macro penalty ($\Delta\text{MAE} = 0.0038$), although its interval spans
 603 zero. DM changes by -0.0009 , while HDM and DD remain near zero. In DM
 604 and HDM, every disturbed current sample enters the cumulative charge state
 605 Q_c , but positive and negative noise increments partly cancel during integration.
 606 The integration therefore smooths rapid zero-mean fluctuations in the resulting
 607 SOC trajectory even though no individual sample is ignored. HECM processes
 608 disturbed current directly in both state propagation and the voltage-prediction
 609 path, which provides a plausible transmission mechanism for its larger mean re-
 610 sponse. DD also receives the disturbed raw current directly and, unlike DM and
 611 HDM, additionally receives dI/dt at every recurrent step. These direct channels
 612 explain the visibly larger short-term DD output variation in Figure 7(d). The
 613 rolling multichannel context distributes this response over the input sequence.

At the tested level, the local transmission does not develop into a persistent global bias, so DD remains near zero in the window-wide ΔMAE . These explanations remain conditional on the reported uncertainty because the HECM interval includes zero. Voltage noise ($\sigma_U = 0.01$ V) produces a 0.0014 penalty for DM and 0.0013 for HDM, while HECM and DD remain below 0.0003. The small DM and HDM voltage response can arise through voltage-dependent full-charge anchoring rather than their continuous current-integration update. Temperature noise is neutral for DM and small for the SOH-aware models because temperature enters those pipelines through learned or parameter-dependent pathways rather than the direct charge update. Figure 7 shows the local transmission of current noise, while Figure 8 reports repeated six-cell effects with hierarchical uncertainty.

Taken together, the repeated six-cell results show that zero-mean sensor noise causes only small changes in window-wide accuracy for most model-channel combinations. The local current-noise analysis further demonstrates that window-wide ΔMAE should be interpreted together with short-term output transmission because temporal fluctuations may be only weakly represented by an aggregate error metric.

4.5. Initialization error and recovery

Initialization performance uses the paired criterion defined above. The protocol maps the same -0.10 SOC-equivalent mismatch to the explicit DM/HDM Coulomb-counting state, the HECM EKF SOC state, and a Q_c offset of $-0.10C_{\text{nom}}$ for DD. Figure 9 reports the resulting paired trajectory differences from the common evaluation start, while recovery time remains referenced to the intervention. HECM provides the fastest persistent recovery, with a cell-macro mean of 1.20 h, followed by DD at 2.60 h. DD enters the recovery band temporarily in several runs but has a higher relapse fraction than HECM (0.78 compared with 0.17). DM and HDM generally do not recover persistently within the observation horizon, and both have a persistent-censor fraction of 0.89. HECM also has the smallest excess-error area at 0.078 SOC h, compared with 0.116 SOC h

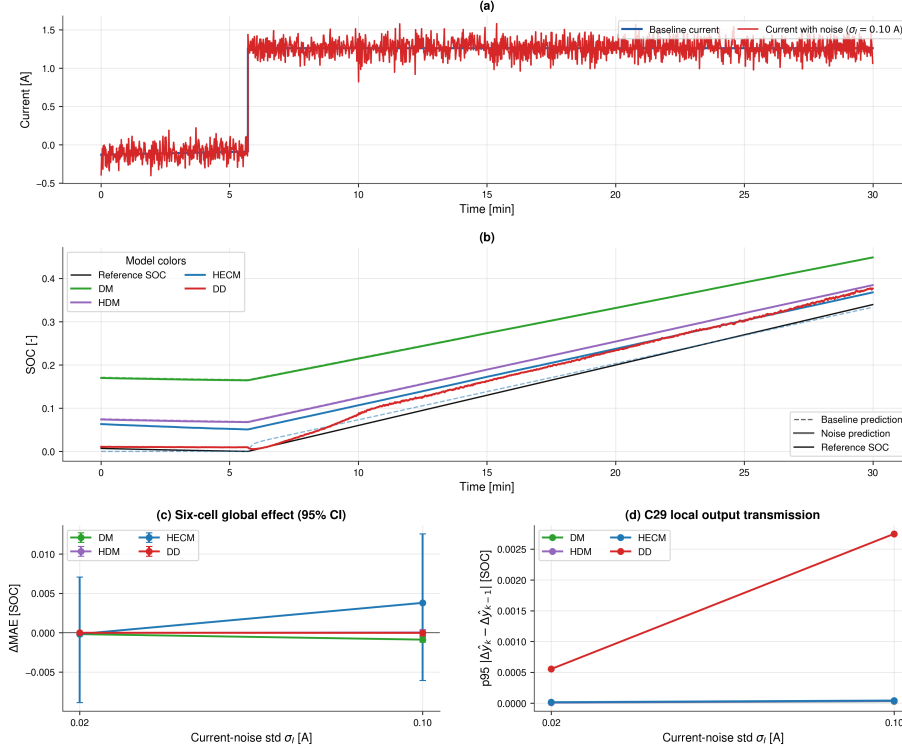


Figure 7: Noise sensitivity under additive current noise. (a) Baseline and noisy current in a reset-free 30 min example from the high-load test cell. (b) Matched baseline and noisy SOC trajectories in the same window. (c) Cell-macro ΔMAE with hierarchical 95% confidence intervals across the holdout test set. (d) Local transmission in the same example, measured as the 95th percentile of the change in noise-induced output deviation between consecutive samples. DD shows the largest local transmission because raw current and its derivative enter the recurrent input directly, whereas the integration in DM and HDM smooths zero-mean sample-level fluctuations. The local panels illustrate the mechanism, while panel (c) provides the test-set comparison.

644 for DD. The recovery pattern follows the available correction mechanisms. DM
 645 and HDM propagate the imposed state offset until a physical charge anchor
 646 is reached because neither method has continuous output feedback. HECM
 647 uses the voltage residual to correct its explicit SOC state and can therefore
 648 re-anchor persistently. DD does not propagate SOC as an isolated state. Its
 649 multichannel rolling input allows the prediction to move away from the dis-

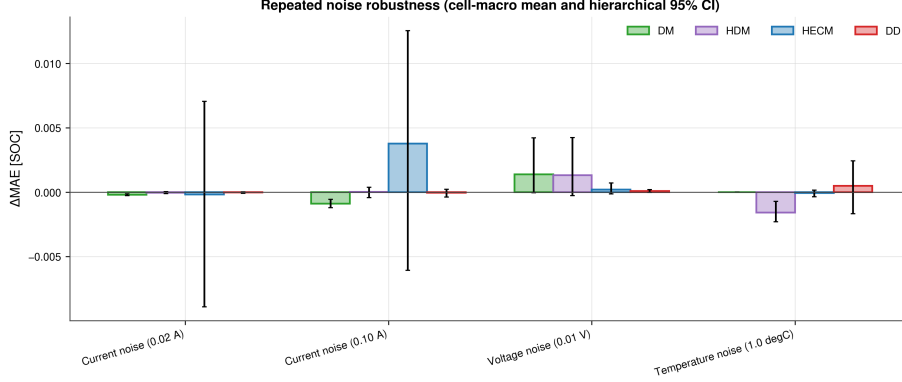


Figure 8: Repeated noise robustness across the six-cell holdout test set. Bars show cell-macro ΔMAE and error bars show hierarchical 95% confidence intervals with random seeds nested within cells. HECM is most affected by high current noise, while most other channel-model combinations remain close to baseline relative to the between-cell uncertainty.

650 turbed cumulative-charge feature, although the interaction of that feature with
 651 the changing recurrent context explains why entry into the recovery band can
 652 be followed by relapse. Together, the recovery time, relapse behavior, censoring,
 653 and excess-error area identify HECM as the strongest recovery implementation
 654 under this protocol.

655 4.6. Missing samples and signal integrity

656 Figure 10 compares the six-cell effects of periodic and random sample loss
 657 with those of timing jitter. Sample loss is substantially more consequential than
 658 timing jitter. Periodic freezing of every 50th sample increases MAE by 0.0072,
 659 0.0065, 0.0028, and 0.0011 for DM, HDM, HECM, and DD, respectively. Ran-
 660 dom 2% loss produces corresponding increases of 0.0080, 0.0081, 0.0039, and
 661 0.0026. By contrast, even the ± 0.9 s jitter case remains within approximately
 662 6×10^{-4} SOC of baseline for every model, with confidence intervals generally
 663 spanning zero. DD therefore has the smallest sample-loss penalty, while DM
 664 and HDM retain their structural dependence on uninterrupted charge integra-
 665 tion. A missing or frozen current sample removes or distorts charge information
 666 that an integration-based estimator cannot reconstruct retrospectively. HECM

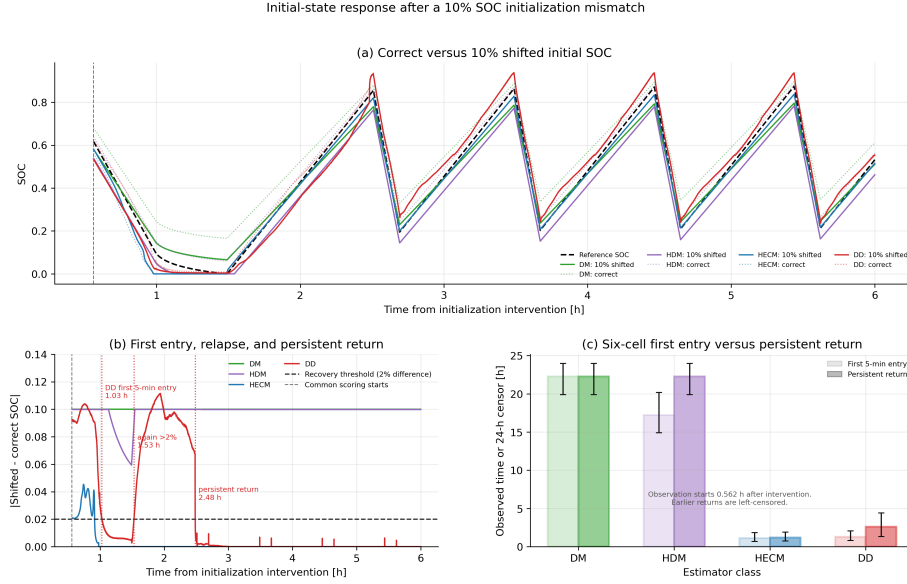


Figure 9: Response to a 10% SOC-equivalent initialization mismatch. (a) Correctly and incorrectly initialized trajectories for a representative mid-life pair from the high-load holdout trajectory. (b) Absolute paired differences over the first 6 h. The DD trajectory enters the 2% band for 5 min, leaves it again, and only later returns persistently. The vertical gray marker denotes the common evaluation start after all estimators provide valid output. (c) Observed first qualifying entry and persistent recovery-or-censor time across the holdout test set with hierarchical 95% confidence intervals. Endpoints already satisfied at the common start are left-censored.

667 can reduce part of this error through voltage feedback, while DD can use the
 668 surrounding multichannel sequence rather than relying on the affected sample
 669 alone. Timing jitter preserves the measurements themselves, and each imple-
 670 mentation reconstructs its time-dependent quantities from the disturbed sam-
 671 pling intervals. This explains why timing perturbation is less consequential than
 672 information loss under the tested protocol.

673 The one-hour burst dropout creates a distinct availability problem. Against
 674 the duration-matched 48-h baseline, its macro ΔMAE is largest for HECM
 675 (0.0239, 95% CI 0.0155–0.0366), followed by HDM (0.0209, 0.0098–0.0313) and
 676 DM (0.0050, -0.0029 –0.0138). DD changes by -0.0093 (-0.0288 –0.0099), re-

677 flecting finite-window compensation between its baseline and frozen-input tra-
 678 jectories. The DD interval includes zero, so the negative mean does not demon-
 679 strate an accuracy improvement. It shows that no persistent global degradation
 680 is resolved under this protocol. The interval is placed where the absolute un-
 681 observed net charge is largest among starts that retain the required pre- and
 682 post-gap context. Figure 11 compares the disturbed output of the represen-
 683 tative high-load test cell with its duration-matched no-dropout trajectory and
 684 reports the global holdout-test effect. DM and HDM retain the corrupted charge
 685 balance until a physical anchor because the missing interval cannot be recon-
 686 structed by integration alone. HECM receives measurements again after the
 687 gap, but the flat LFP voltage relation limits how strongly its observer can infer
 688 the missing charge and correct the accumulated state deviation. DD progres-
 689 sively removes frozen raw and derivative samples from its finite rolling window
 690 after measurements resume. Its voltage, temperature, current, and temporal
 691 context can then reduce dependence on the corrupted cumulative-charge fea-
 692 ture. This mechanism is consistent with its smaller persistent effect, while the
 693 finite-window confidence interval prevents a stronger claim. Dropout contributes
 694 to the robustness dimension, while the recovery dimension uses the canonical
 695 paired initialization experiment.

696 4.7. *Spike sensitivity*

697 Single voltage spikes demonstrate why local and global metrics must be
 698 reported together. Full-window ΔMAE remains below 4×10^{-4} SOC for every
 699 estimator and is effectively zero for DM, HDM, and DD. Event-aligned analysis,
 700 however, identifies a reproducible DD transient: its mean spike-sample penalty
 701 across cells is approximately 7.7×10^{-3} SOC, compared with near-zero direct
 702 penalties for the other representatives. Figures 12 and 13 further show that the
 703 DD response varies across cells and SOH states and then decays after the event.
 704 DM and HDM do not use voltage as a continuous SOC-correction signal, so an
 705 isolated spike away from the full-charge condition has little direct effect. HECM
 706 processes voltage through its observer residual, which filters a single inconsistent

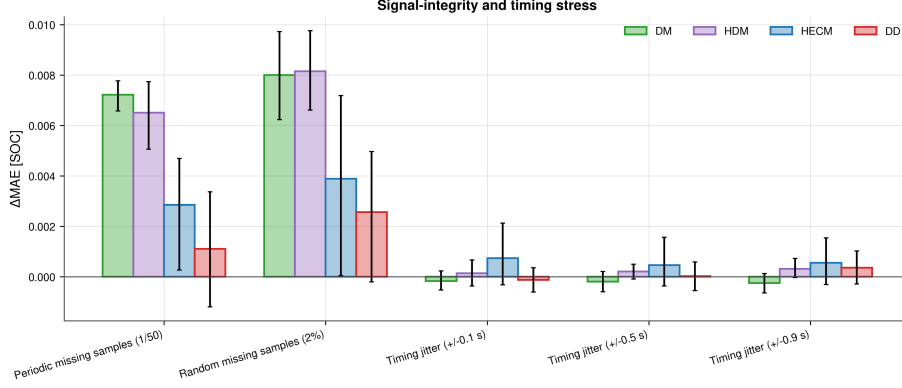


Figure 10: Six-cell signal-integrity and timing stress. Bars show cell-macro ΔMAE and hierarchical 95% confidence intervals for periodic and random sample loss and three timing-jitter levels. Sample loss clearly dominates timing jitter under the tested protocol.

707 measurement through the model and covariance weighting. DD receives the
 708 impulse simultaneously through absolute voltage and dU/dt , and the disturbed
 709 values enter its rolling recurrent context. This provides a plausible mechanism
 710 for the larger but short-lived DD response. A channel-removal ablation would be
 711 required to separate the individual contributions of the absolute and derivative
 712 inputs.

713 4.8. ADC quantization

714 Figure 14 compares representative raw and quantized current and voltage
 715 signals and summarizes the resulting six-cell ΔMAE . The tested ADC steps
 716 ($\Delta I = 0.01\text{ A}$, $\Delta U = 0.005\text{ V}$, and $\Delta T = 0.5^\circ\text{C}$) preserve the local signal
 717 shape but affect the tested representatives differently. DM and HDM increase
 718 by 0.0038 and 0.0035 MAE, respectively. HECM changes by -0.0018 and DD
 719 by $+0.0002$. All four confidence intervals overlap zero. The small DM and
 720 HDM response is consistent with repeated current-rounding errors entering their
 721 cumulative charge states. HECM combines the quantized current with voltage-
 722 residual correction, while DD embeds all quantized channels in a multichannel
 723 temporal context. Their signed changes remain within the uncertainty range
 724 and must not be interpreted as physical improvement or exact cancellation.

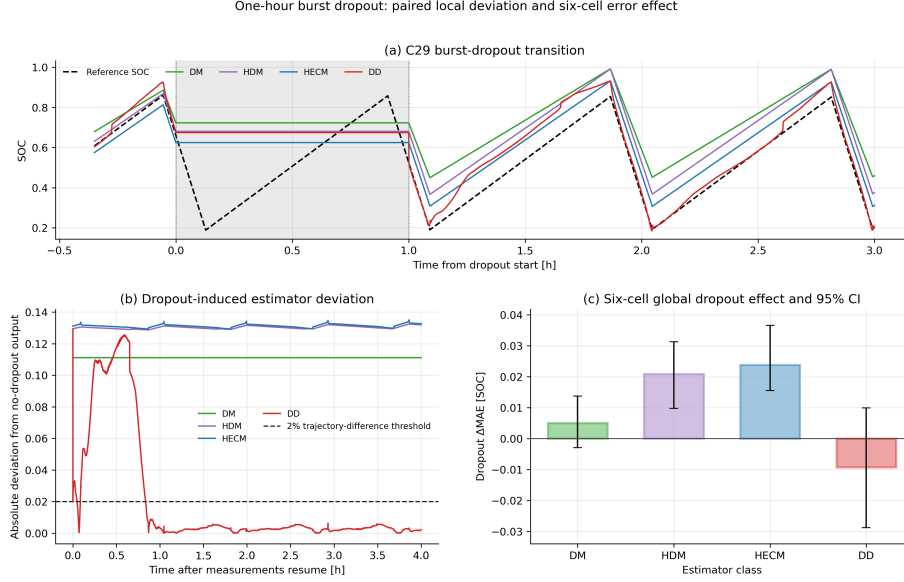


Figure 11: One-hour burst-dropout transition. (a) Estimator outputs around the shaded missing-input interval. (b) Absolute deviation from the paired duration-matched no-dropout output after measurements resume, with the 2% trajectory-difference line shown for context. (c) Six-cell global Δ MAE with hierarchical 95% confidence intervals.

At the selected coarse resolution, quantization produces smaller penalties than current gain error, sample loss, and burst dropout, while the integration-based representatives retain a measurable response.

4.9. Diagnostic comparison with ground-truth SOH

The primary comparison supplies the same causally estimated LSTM-SOH trace to HDM, HECM, and DD. This estimated trace represents the health information available to the complete estimator pipelines during online operation. The diagnostic comparison replaces it with the dataset reference SOH, which is the ground-truth SOH derived from capacity checkups and interpolated between the measured anchors. Its purpose is to identify whether observed SOC errors originate predominantly within the SOC branch or are inherited through the estimated health input.

Reference SOH reduces baseline MAE by 0.0370 for HDM and 0.0166 for

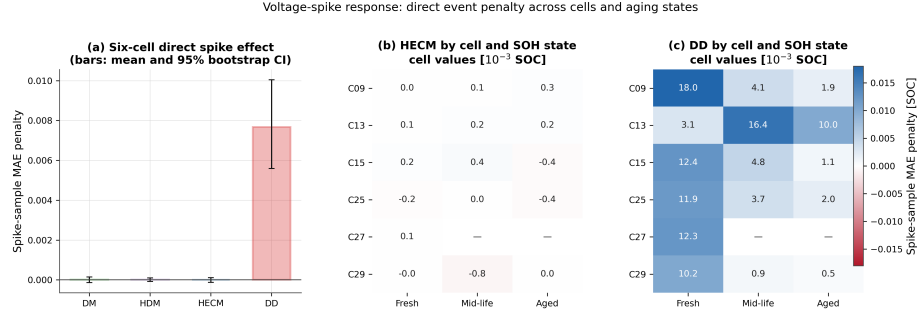


Figure 12: Voltage-spike sensitivity across six cells. (a) Mean event-sample MAE penalty with hierarchical 95% confidence intervals. (b,c) Cell- and SOH-state-resolved penalties for HECM and DD, respectively. Absent cell/state combinations are left blank. The event-aligned metric exposes the local DD response that is obscured in full-window Δ MAE.

HECM, while DD MAE increases by 0.0049 on average. The pronounced HDM improvement is consistent with its direct use of SOH to scale effective capacity. HECM also benefits because SOH enters both capacity scaling and lookup-based parametrization. DD instead uses SOH as one learned feature among several interacting inputs. Replacing this single feature with its ground truth therefore does not impose a monotonic relationship between SOH accuracy and the resulting SOC error.

The same model-specific direction persists for most noise, sample-loss, and dropout cases. At 3% current gain error, reference SOH improves HDM and HECM and changes DD by +0.0091. For temperature noise, the reference-minus-LSTM difference changes from its baseline value by 0.0016 for HDM and less than 10^{-4} for HECM and DD. The incremental temperature-noise response is therefore largely independent of the SOH estimator. These findings show that SOH-input quality explains part of the baseline differences, especially for HDM and HECM, but does not account for every disturbance-specific response. The LSTM-SOH runs consequently remain the primary deployment-representative results, while the ground-truth SOH runs provide the diagnostic attribution.

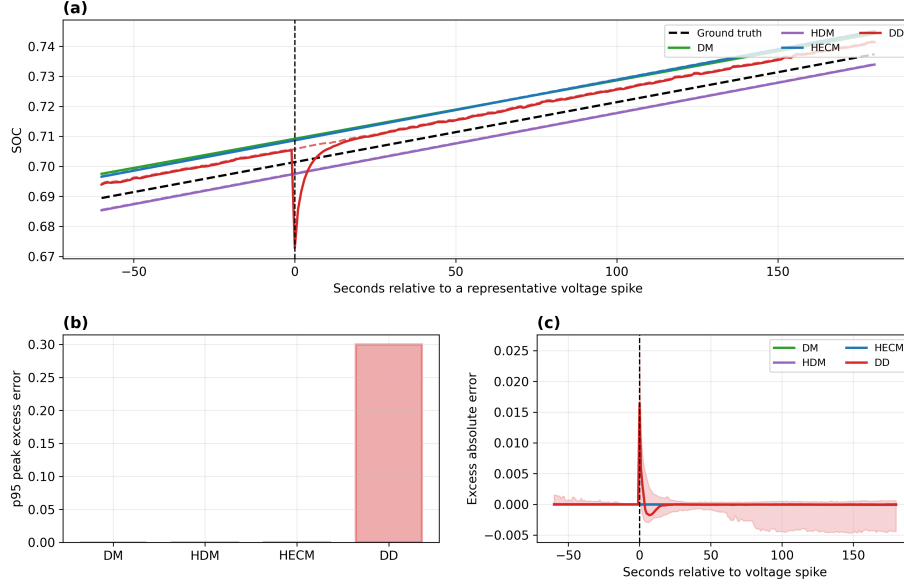


Figure 13: Event-aligned voltage-spike response. (a) Representative SOC trajectories around the injected spike, with the dashed black line showing dataset ground-truth SOC. (b) Model-wise 95th percentile peak excess error. (c) Mean excess absolute error aligned across repeated spikes, with the DD uncertainty envelope. The local response decays rapidly and is therefore weakly represented by full-window MAE.

4.10. HECM lookup-table sensitivity

Within the tested local perturbation range, lookup-table uncertainty is not the dominant source of the observed HECM measurement-disturbance penalties. Across all 22 measurement and signal-integrity subcases, the largest absolute lookup-disturbance interaction is 0.006116 MAE. It occurs within the additive current-offset test, but its cell-bootstrap interval includes zero. This interaction is small relative to the HECM current-offset penalty of 0.1763 MAE obtained with the nominal lookup. The large offset response is therefore attributable primarily to the systematic measurement error and not to the tested variations of the parameter surfaces.

The lookup tables nevertheless remain relevant to baseline calibration and recovery. Five of the six lookup perturbations retain mean persistent recovery-

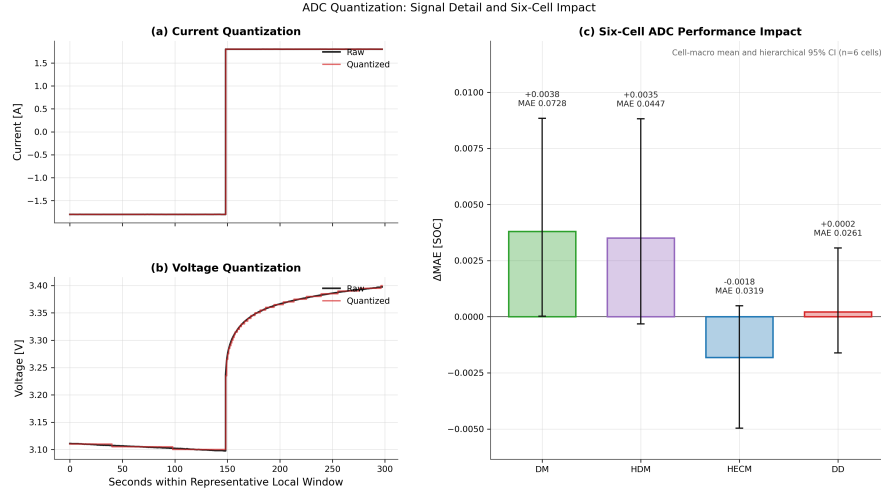


Figure 14: ADC quantization. (a,b) Representative raw and quantized current and voltage. (c) Six-cell Δ MAE with hierarchical 95% confidence intervals and resulting quantized-signal MAE annotations.

or-censor times between 1.18 and 1.41 h, close to the nominal value of 1.20 h. Under the remaining resistance perturbation, one low-load holdout cell stays outside the recovery band for the complete observation horizon and raises the equal-weight mean to 5.10 h. The analysis therefore answers the primary disturbance-attribution question negatively while identifying recovery as a separate calibration-sensitive boundary. Figure B.26 and Table B.5 provide the complete Appendix results.

4.11. Robustness synthesis across representatives

Figure 15 condenses 18 measurement-corruption and signal-integrity rows and shows how degraded-input behavior changes the nominal-accuracy picture. The additive current offset creates the largest global penalties for DM, HDM, and HECM, while DD remains less affected. Current-gain sensitivity increases with magnitude for every representative in the matched signed sweep. Periodic and random sample loss particularly affect DM and HDM, and the one-hour dropout produces the largest non-offset point penalty for HECM. Timing jitter

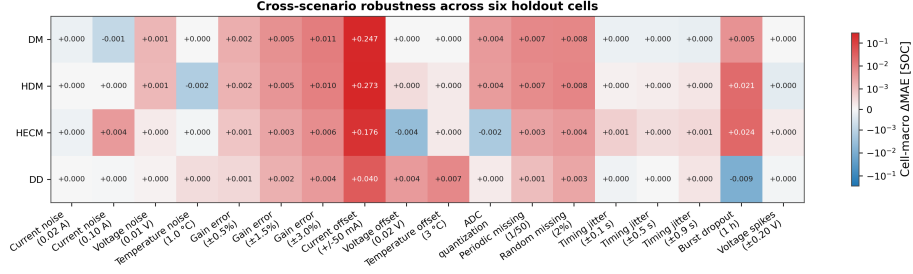


Figure 15: Cell-macro Δ MAE across 18 measurement-corruption and signal-integrity rows on the six-cell holdout test set. Current-gain and additive current-offset rows use adverse directions from matched signed sweeps. A symmetric logarithmic color scale retains visual resolution for smaller effects despite the larger current-offset penalties. Positive values denote degradation relative to each model’s paired baseline, while small negative values arise from finite-window signed compensation. The initialization-mismatch response is evaluated separately in Figure 9.

remains nearly neutral. Voltage spikes reveal a distinct DD transient through event alignment despite their small full-window MAE. Figure 16 complements this scenario-level comparison by combining relative accuracy, family-balanced robustness, and recovery and by showing how alternative priority profiles change the overall assessment. Together, both figures provide the evidence required for deployment-oriented estimator selection.

Under the family-balanced definition, the robustness scores are 0.452 for DM, 0.351 for HDM, 0.360 for HECM, and 0.967 for DD. Equal weighting per evaluated scenario yields 0.549, 0.476, 0.552, and 0.793, while the highest-level-plus-offset variant yields 0.573, 0.469, 0.562, and 0.693. DD leads all three aggregations. The changing score distances and the changing order of DM and HECM demonstrate the influence of the aggregation rule, while the scenario-level results identify the disturbance mechanisms behind these relative summaries.

4.12. Embedded performance benchmark

The STM32 experiment first verifies numerical equivalence before comparing speed or memory. Across the six nominal cell sequences, mean hardware–software MAE remains below 4×10^{-4} percentage points for every SOC core

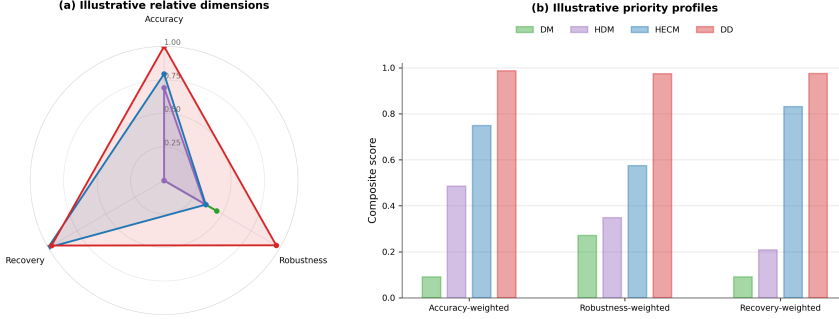


Figure 16: Decision-oriented synthesis for the four tested representatives. (a) Relative min-max-normalized accuracy, family-balanced robustness across eight evaluated disturbance families, and observed persistent paired recovery. Current gain and additive current offset use adverse directions from their matched signed sweeps. (b) Composite scores under three 60/20/20 priority profiles. The profiles are interpreted together with the scenario-level evidence in Figure 15.

799 (Figure 17). These differences are orders of magnitude below the estimator-
800 to-dataset errors and show that the float32 C implementations reproduce their
801 software references sufficiently closely for deployment measurements.

802 The isolated DM and HDM SOC updates both have sub-microsecond me-
803 dian latency (0.171 and 0.165 μs), HECM requires 23.9 μs , and continuous-state
804 DD requires 424.9 μs (Figure 18). The latency order follows the computation
805 performed per update. DM and HDM require only a small number of scalar
806 arithmetic operations. HECM additionally performs parameter lookup and ob-
807 server matrix operations. DD evaluates recurrent matrix products and its MLP
808 output head. At the nominal 1-Hz update rate, these medians occupy less
809 than 0.043% of one sampling period. This deadline share describes the iso-
810 lated estimator task and provides the execution margin available for integration
811 with other BMS functions. The DD value in the cross-model comparison is the
812 continuous-state deployment variant. Exact 2024-sample rolling-window seman-
813 tics are evaluated separately because they recompute the full sequence at every

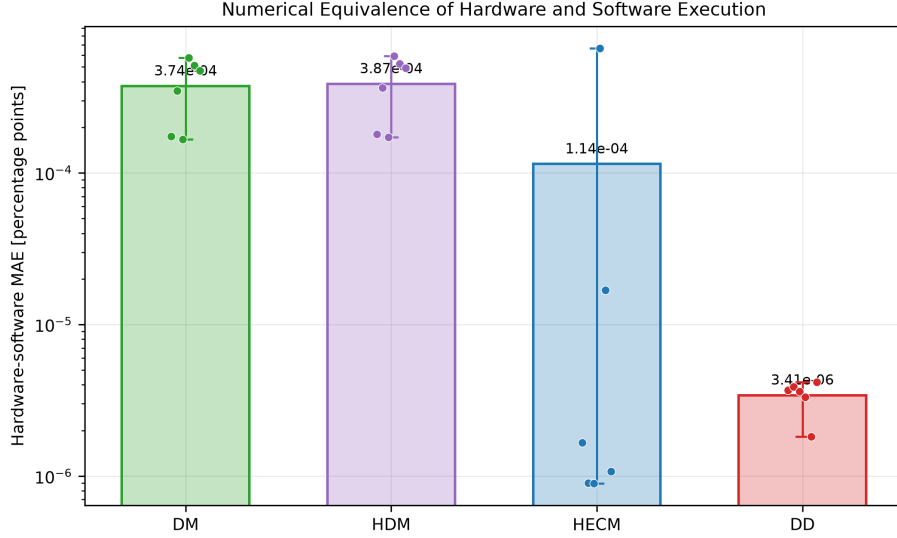


Figure 17: Numerical equivalence of STM32 and software execution over six cell sequences. Bars show mean hardware–software MAE in percentage points, dots show cells, and whiskers show the cell range. The logarithmic scale emphasizes that all implementation differences are negligible relative to model error.

sample.

For the cross-model embedded performance comparison, DD denotes the continuous-state deployment variant used in the latency analysis. The compiled DM, HDM, HECM, and DD firmware occupies 27.0, 27.1, 470.1, and 115.9 KiB of flash, respectively. The corresponding measured peak runtime RAM is 2.4, 2.4, 2.5, and 4.1 KiB (Figure 19). These peak values combine static storage, dynamic allocation, and call-stack demand. HECM flash occupancy is dominated by its parameter surfaces, whereas the DD parameter storage accounts for most of its flash demand. The measurements describe the isolated SOC firmware with the shared SOH trace provided externally and leave resource margin for additional BMS tasks.

Figure 20 makes the DD deployment trade-off explicit, while Figure A.24 provides the corresponding latency distributions over all six hardware cell sequences. Exact rolling-window inference preserves the trained software seman-

On-Device Inference-Latency Distributions Across Six Cells

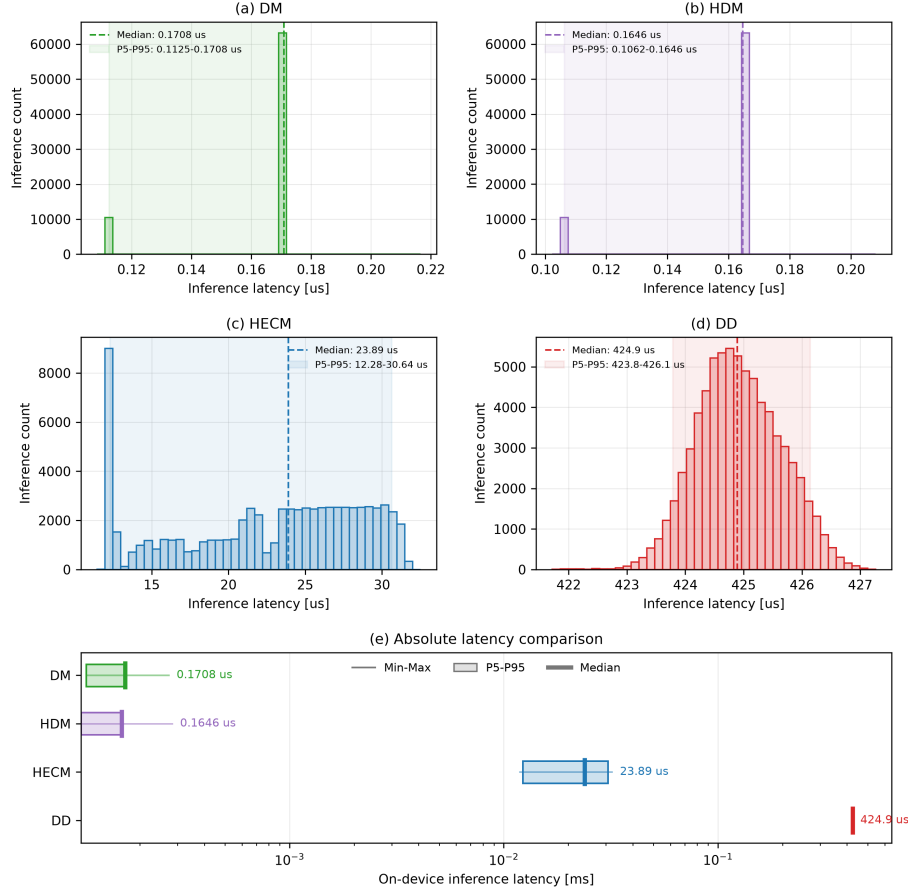


Figure 18: On-device SOC-core latency distributions across six cells and three replay rounds. Panels (a–d) show per-inference distributions. Panel (e) compares minimum, 5th percentile, median, 95th percentile, and maximum on a logarithmic scale. DD denotes continuous-state inference in this cross-model latency comparison.

tics but requires roughly 724 ms median inference time, or 72.4% of the one-second period, and 67.3 KiB peak runtime RAM because the complete 2024-sample sequence is recomputed for every output. Continuous-state inference evaluates only the new recurrent step and reduces latency to approximately 0.425 ms and peak RAM to 4.1 KiB while retaining similar error on the six

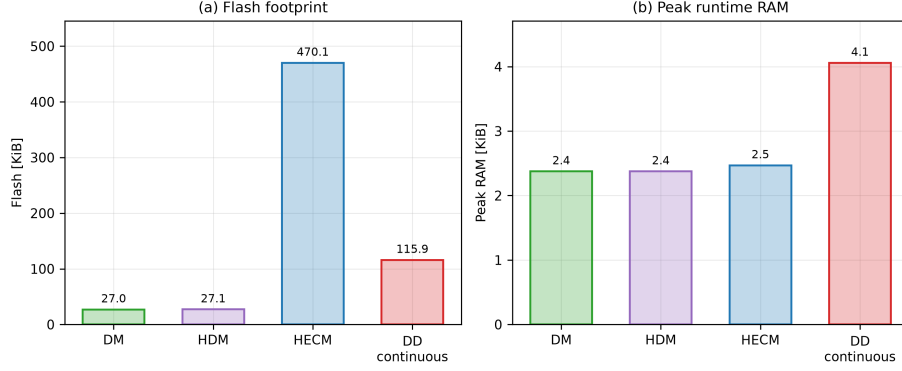


Figure 19: Embedded memory requirements of the isolated STM32 SOC implementations. (a) Compiled flash occupancy. (b) Peak runtime RAM measured during a representative holdout-test replay, combining statically allocated variable storage with maximum dynamic-memory allocation and call-stack use. DD denotes continuous-state execution, matching Figure 18. Exact rolling-window execution is compared separately in Figure 20.

nominal sequences. Periodically resetting that state has a comparable 4.1 KiB peak but creates much larger transient errors, with a maximum SOC error near 28 percentage points. These transients arise because each reset removes the temporal context accumulated before the reset, whereas continuous forwarding preserves it. Continuous-state execution therefore provides the practical hardware configuration. Its forwarded recurrent state defines a deployment mode that is evaluated separately from the primary rolling-window benchmark.

5. Discussion

5.1. Structural interpretation across estimator classes

The four implementations expose distinct relationships between their update mechanisms and observed disturbance responses. DM has the largest nominal error and reacts strongly to persistent current-gain and current-offset errors as well as missing samples because its current-integration structure lacks voltage-residual correction. Its practical full-charge anchor limits accumulated drift, while the lifecycle experiment shows that the reset restructures the gain-error

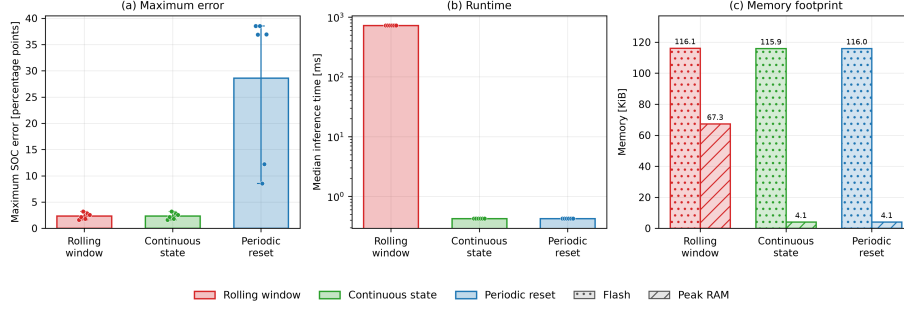


Figure 20: DD inference-mode trade-off on the STM32. The panels compare maximum dataset-SOC error, median latency, compiled flash occupancy, and measured peak runtime RAM for exact rolling-window, continuous-state, and periodically reset recurrent execution. Cell points show that periodic resets, not continuous forwarding, create the dominant accuracy loss.

848 contribution without removing it uniformly. HDM adapts the capacity denom-
849 inator through learned SOH and thereby reduces capacity mismatch, yet it
850 retains the same charge-integration core. Its similar response to systematic cur-
851 rent errors, missing charge information, and ADC quantization follows directly
852 from this shared structure.

853 HECM combines current-driven state propagation with a voltage residual
854 and has the strongest observed recovery profile under the tested initialization
855 mapping. Additive current offset produces its largest global disturbance penalty.
856 Among the remaining cases, high current noise and burst dropout are most
857 consequential because corrupted or frozen inputs affect both state propaga-
858 tion and observer correction. The confidence interval for high current noise in-
859 cludes zero and shows substantial between-cell uncertainty. The supplementary
860 lookup analysis in Appendix Appendix B shows that the largest local lookup-
861 disturbance interaction is 0.00612 MAE, compared with a 0.1763 HECM penalty
862 from the adverse current offset itself. The disturbance response is therefore gov-
863 erned primarily by the corrupted input in this case, although absolute accuracy
864 and observer re-anchoring remain dependent on lookup calibration.

865 DD has the lowest nominal cell-macro error and the smallest adverse penal-

ties under both systematic current-error tests. It is also comparatively resilient to isolated sample loss. Event-aligned voltage spikes expose a recurrent transient that full-window MAE hides. The selected voltage and derivative features and the rolling recurrent context provide a plausible transmission path for this response and identify a concrete target for feature ablation and disturbance-aware training.

The ground-truth SOH comparison demonstrates why the auxiliary state estimator must be considered part of the complete SOC pipeline. Part of the HDM and HECM baseline error is inherited through the estimated SOH input, whereas the DD response reflects a learned interaction between SOH and the remaining features. A more accurate auxiliary input therefore improves a physically defined capacity relation more directly than it improves a multivariate learned mapping. At the same time, the limited change in several paired disturbance penalties shows that the shared SOH estimator is not the dominant cause of every robustness response. This distinction prevents errors of the auxiliary SOH service from being attributed incorrectly to the SOC core and supports evaluation of the complete causal pipeline as the primary configuration.

At class level, these representative results identify transferable mechanisms rather than a universal ranking. Direct-measurement methods provide transparent and deterministic updates with minimal computational and memory demand. Their update structure primarily propagates accumulated charge, however, and offers no continuous mechanism for reconstructing missing charge information or correcting systematic integration errors. Reliable current calibration, uninterrupted sampling, and physical anchoring are therefore central to their robustness. Adding learned SOH information can reduce capacity mismatch in a hybrid direct-measurement method, but it does not by itself add voltage feedback or temporal correction to the underlying integration core.

Model-based ECM observers extend this structure with a physically defined correction mechanism. The voltage residual allows HECM to re-anchor its internal SOC state and is consistent with its strong observed recovery behavior. This correction does not provide the same breadth of measurement history as

897 a learned multichannel sequence representation. Its effectiveness remains con-
 898 strained by the observability of the operating point, the flat OCV-SOC re-
 899 lation of LFP cells, the calibrated parameter surfaces, and disturbances that
 900 affect both current-driven state propagation and observer correction. The data-
 901 driven representative instead combines current, voltage, temperature, derived
 902 quantities, and their temporal development within one recurrent context. This
 903 broader context provides a plausible mechanism for its low nominal error and
 904 its comparatively small penalties under several systematic-error and sample-loss
 905 scenarios. The results therefore support the value of multivariate temporal in-
 906 formation without establishing that every data-driven estimator will outperform
 907 every physical model.

908 Temporal context is not exclusively beneficial. The voltage-spike experiment
 909 shows that a localized disturbance can propagate through selected features and
 910 the recurrent state, while the periodic-reset experiment shows that removing
 911 accumulated context can create large transient errors. The same mechanism
 912 that allows DD to combine information across time can therefore transmit dis-
 913 turbances or become unavailable when its state is handled incorrectly. Robust
 914 data-driven deployment consequently requires representative training coverage,
 915 disturbance-aware evaluation, careful feature selection, and a state-handling
 916 strategy that matches the intended embedded execution mode.

917 5.2. Limitations of nominal accuracy

918 Nominal MAE describes average agreement with the dataset ground truth
 919 under clean inputs. The disturbance and recovery experiments add the DD
 920 voltage-spike transient, HECM re-anchoring behavior, shared-SOH feature cou-
 921 pling, and model-specific responses to random noise, systematic sensor errors,
 922 and missing information. Embedded measurements add latency, memory, and
 923 inference-state semantics. These dimensions create a deployment profile for each
 924 implementation, while the weighting analysis shows how application priorities
 925 reshape their combined scores. Estimator selection can therefore begin with
 926 the relevant fault and recovery requirements and then incorporate the available

927 compute, memory, and state-service architecture.

928 5.3. Implications for BMS deployment

929 For BMS deployment, estimator classes should be selected according to
930 the dominant sensing risks, recovery requirements, and hardware constraints.
931 Direct-measurement methods are attractive when transparency, deterministic
932 execution, and minimal memory are primary requirements, provided that cur-
933 rent calibration, reliable sampling, and physical anchoring can be ensured. Hy-
934 brid methods can add adaptive health information or learned correction to an
935 existing physical core. Their benefit and their remaining sensitivities depend on
936 which part of that core is modified. Model-based ECM observers provide feed-
937 back correction and stronger re-anchoring potential at the cost of parameteriza-
938 tion effort and larger storage requirements. Data-driven methods provide broad
939 flexibility for representing nonlinear dependencies and temporal context, but
940 require representative training data and explicit validation of feature-mediated
941 transients and recurrent-state behavior.

942 The STM32 results show that hardware resources do not constitute a fun-
943 damental deployment barrier for any of the four isolated SOC cores on the
944 selected microcontroller. Every implementation fits into the available memory
945 and completes its update within the one-second sampling interval. This re-
946 sult does not make computational constraints irrelevant, because the measured
947 resource profiles differ substantially. DM and HDM provide the smallest and
948 fastest implementations. HECM remains fast but requires 470.1 KiB of flash,
949 which is dominated by its parameter surfaces. The continuous-state DD im-
950 plementation requires more computation and runtime memory than the other
951 representatives, yet its median latency of approximately 0.425 ms and peak run-
952 time RAM of 4.1 KiB demonstrate that recurrent neural estimation is feasible
953 within the tested embedded environment.

954 The comparison of DD execution modes further shows that embedded suit-
955 ability depends on implementation semantics as well as model architecture. Ex-
956 act rolling-window execution requires approximately 724 ms and 67.3 KiB peak

runtime RAM because it repeatedly evaluates the complete input sequence. Continuous-state forwarding evaluates only the new recurrent step and retains similar nominal error while using a small fraction of the sampling period. Periodic state resets preserve the low computational cost but produce large transient errors because they remove the temporal information accumulated before each reset. Model selection and embedded optimization must therefore be considered together. A model that is computationally demanding under one execution strategy can become practical under another strategy, provided that the resulting state semantics remain consistent with the required estimation behavior. The firmware measurements cover flash, maximum call-stack use, dynamic-memory demand, and numerical equivalence. Integration of the shared SOH-LSTM, energy measurement, pack communication, and concurrent BMS scheduling forms the next system-level validation stage. Table 3 relates the measured estimator profiles to concrete deployment priorities.

Figure 16 complements the recommendation map by combining the measured accuracy, robustness, and recovery dimensions into relative priority profiles. The three profiles assign 60% to their named dimension and 20% to each remaining dimension. Family-balanced, equal-scenario, and high-severity aggregations produce different score distances and change the HDM-HECM order. This sensitivity turns the score into a configurable decision aid, while the scenario heatmap identifies the disturbance-specific strengths and weaknesses behind each profile.

5.4. Limitations and future extensions

The conclusions are grounded in four specified implementations and the six-cell holdout test set from one controlled LFP aging dataset. Alternative reset logic, ECM structures, neural architectures, features, and training data can produce different estimator profiles. The low- and middle-load strata contain multiple cells, whereas the high-load stratum contains one exploratory trajectory. One low-load test trajectory also provides no mid-life or aged window. Figure A.21 makes these coverage limits explicit. The reported confidence in-

Table 3: Decision-oriented recommendation map derived from the benchmark results. The table identifies the strongest observed representative under one dominant deployment priority and the stated protocol boundaries.

Primary deployment priority and selected representative	Main supporting benchmark evidence	Main trade-off to consider
Lowest nominal SOC error: DD	Lowest six-cell macro MAE (0.0258) and RMSE (0.0300)	Rolling-window execution is the most computationally expensive implementation
Strongest observed recovery from initialization mismatch: HECM	Lowest observed persistent-recovery-or-censor time and substantially fewer first-entry relapses than DD	Recovery is compared only after all estimators provide valid output
Lowest adverse systematic current-error penalties: DD	Matched signed tests give Δ MAE 0.0038 at 3% gain error and 0.0401 at 50 mA offset, below the other representatives	Temperature-offset and local voltage-spike sensitivity remain visible
Smallest observed periodic/random sample-loss penalty: DD	Δ MAE 0.0011/0.0026	The duration-matched dropout effect is non-positive but highly variable across cells
Lowest deterministic compute cost: DM/HDM	Sub-microsecond median SOC-core latency and approximately 27 KiB flash	Integration structure is sensitive to gain error and missing charge information
Practical fast neural deployment: DD continuous state	Approximately 0.425 ms median and 4.1 KiB peak runtime RAM	Different recurrent semantics from the primary rolling-window benchmark

987 tervals therefore quantify the available between-cell variation within this design.
988 Transfer to other chemistries, cell formats, pack configurations, thermal envi-
989 ronments, and field duty cycles requires corresponding cell-disjoint validation.
990 Nominal accuracy is referenced to one consistently reconstructed offline SOC
991 ground truth based on Coulomb counting and voltage anchors.

992 The primary cross-model campaign concentrates on the BMS measurement
993 chain, signal availability and timing, and deployment-accessible initialization

994 states. The systematic current-error sweep covers matched signs at fixed gain
 995 and offset magnitudes, but not time-varying calibration drift or interacting sen-
 996 sor errors. The separate HECM lookup analysis crosses local resistance, time-
 997 constant, and OCV deviations with the complete set of tested disturbance sub-
 998 cases and paired initialization recovery. These one-at-a-time perturbations do
 999 not define probability distributions for lookup error and do not cover combined
 1000 parameter deviations. Broader parameter mismatch, pack-level interactions,
 1001 and supervisory BMS faults form a separate validation layer. The initialization
 1002 experiment maps an SOC-equivalent offset into each estimator’s available state
 1003 and compares recovery once all estimators provide valid output. The result-
 1004 ing times characterize operational re-anchoring under class-specific interventions
 1005 rather than identical latent-state convergence. The continuous high-load hold-
 1006 out replay complements the test-set windows with a mechanism-focused lifecycle
 1007 view. The hourly SOH service represents gradual capacity fade, and channel-
 1008 removal experiments would further isolate the DD voltage-spike transmission
 1009 path. HECM uses HPPC-derived parameter surfaces identified from the model-
 1010 development sets and held fixed throughout evaluation. The sensitivity analysis
 1011 therefore characterizes local disturbance interactions around this calibrated im-
 1012 plementation rather than arbitrary lookup-table error or every possible HECM
 1013 design.

1014 The hardware profile covers the isolated SOC firmware with SOH supplied
 1015 as an external causal trace. Runtime RAM is measured during a representative
 1016 holdout-test replay, including complete rolling-window DD inference. The next
 1017 hardware stage combines the SOC and SOH services, measures energy demand,
 1018 and evaluates estimator scheduling alongside communication and supervisory
 1019 BMS tasks. Broader data campaigns can then extend the disturbance protocol
 1020 to interacting faults, additional operating environments, and pack-level closed-
 1021 loop behavior.

1022 6. Conclusion

1023 This study uses representative implementations from the direct-measurement,
1024 hybrid, model-based, and data-driven SOC-estimation families to establish a
1025 common robustness and embedded-performance benchmark. Each implementa-
1026 tion was selected to expose a characteristic update or correction mechanism un-
1027 der the same causal protocol. The resulting evidence connects these mechanisms
1028 with the strengths and weaknesses that become relevant when an estimator is
1029 transferred from nominal software evaluation to practical BMS operation.

1030 For direct-measurement methods, the benchmark confirms the advantages of
1031 transparent operation, deterministic execution, and very low resource demand.
1032 It also shows why reliable current sensing, uninterrupted charge information,
1033 and physical anchoring are central to this family. Hybridization can improve an
1034 existing estimator by adding learned health information or adaptive correction,
1035 but its effect depends on the part of the underlying method that is modified.
1036 Capacity adaptation improves the treatment of ageing without automatically
1037 removing the disturbance sensitivities of a current-integration core. Model-
1038 based ECM observers add physically interpretable voltage-feedback correction
1039 and strong re-anchoring capability. These benefits are accompanied by calibra-
1040 tion effort, parameter storage, and dependence on the validity of the physical
1041 model over the intended operating range.

1042 The data-driven family shows particularly strong potential for future BMS
1043 applications. The recurrent representative combines the lowest nominal error
1044 with strong results across several disturbance families, which demonstrates the
1045 value of learning nonlinear and temporal relationships from multichannel battery
1046 data. The hardware benchmark also confirms that this capability is compati-
1047 ble with embedded execution. All tested SOC cores meet the one-second sam-
1048 pling deadline, and the continuous-state neural implementation achieves approx-
1049 imately 0.425 ms median latency with 4.1 KiB peak runtime RAM. Compact
1050 recurrent state estimation is therefore feasible on the selected microcontroller
1051 and is not restricted to offline computation. Representative training coverage,

1052 disturbance-aware evaluation, and careful handling of input features and recur-
1053 rent state remain essential for exploiting this potential reliably.

1054 The broader consequence for all estimator families is that nominal MAE
1055 alone cannot establish suitability for BMS deployment. Accuracy must be evalu-
1056 ated together with disturbance-specific robustness, recovery behavior, numerical
1057 equivalence, inference latency, flash occupancy, and runtime RAM. This multidimensional benchmark does more than compare four implementations. It reveals
1058 the mechanisms that determine class-specific behavior and provides a practical
1059 basis for improving a selected approach through sensing design, physical-model
1060 adaptation, disturbance-aware neural training, or embedded optimization. The
1061 same framework can be extended to complete SOC–SOH execution, concurrent
1062 BMS tasks, additional operating environments, and pack-level applications.

1064 **Appendix A. Dataset Split, Holdout Coverage, and Evaluation Pro-** 1065 **TOCOL**

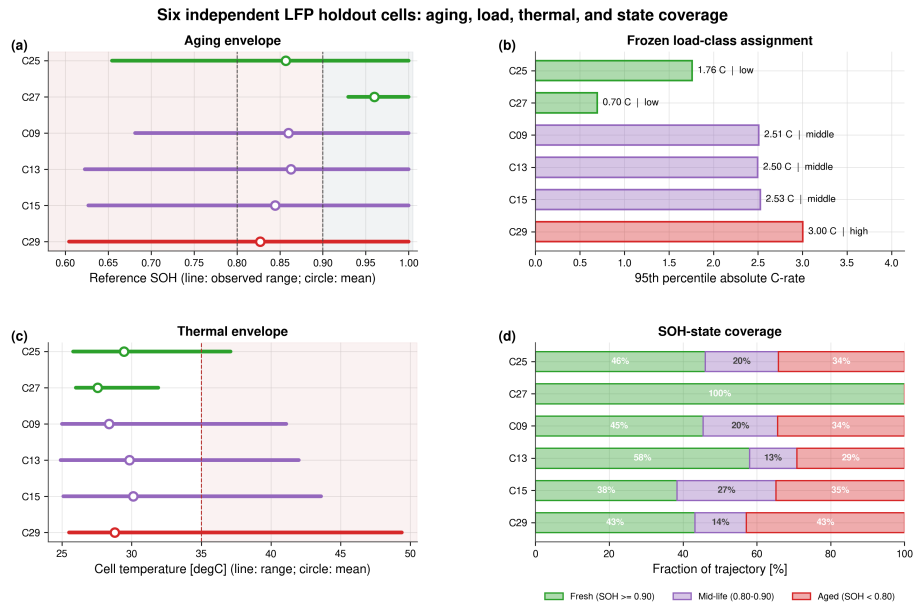


Figure A.21: Quantitative coverage of the six-cell independent holdout test set. Panels show observed SOH, the descriptive load-class assignment, thermal envelope, and full-trajectory SOH-state coverage. The high-load group contains one exploratory trajectory.

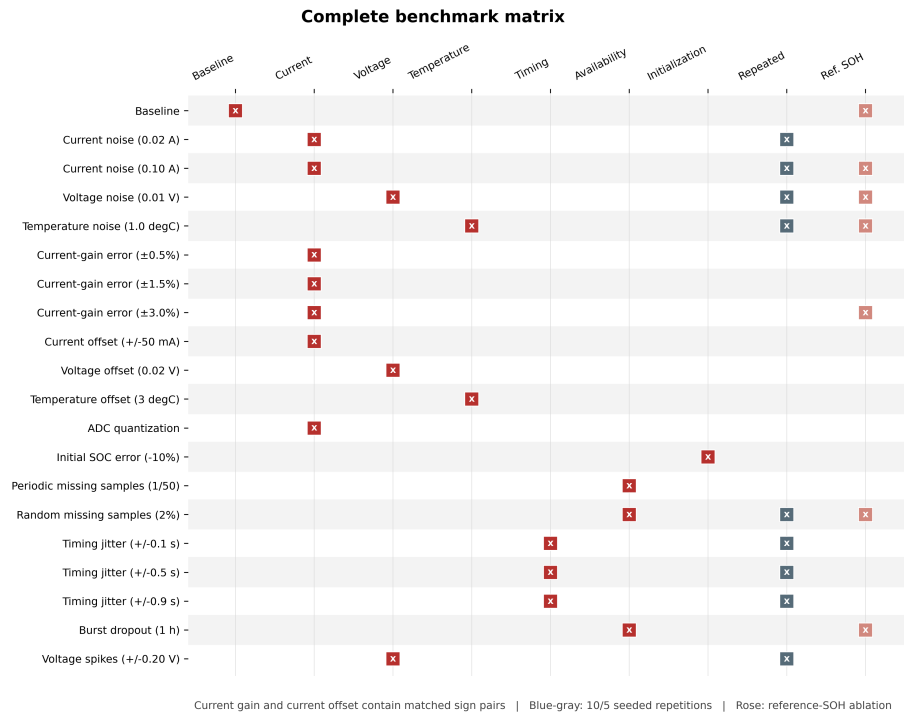


Figure A.22: Complete 20-scenario benchmark matrix. Red markers identify the manipulated measurement, timing, availability, or initialization channel. Blue-gray markers denote repeated stochastic cases. Rose markers identify paired reference-SOH ablations. Each current-gain magnitude and the additive current-offset case contain matched positive and negative sign sublevels.

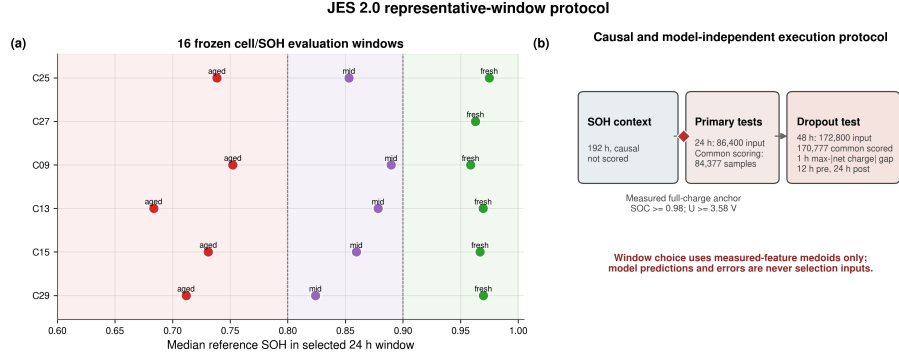


Figure A.23: Representative-window protocol. Sixteen cell/SOH windows are selected using measured-feature medoids. Every estimator is scored over the same interval once the rolling DD input context is complete. Standard windows cover 24 h after the causal SOH context, while the dropout case uses 48 h and places the one-hour gap at the eligible interval with maximum absolute net charge.

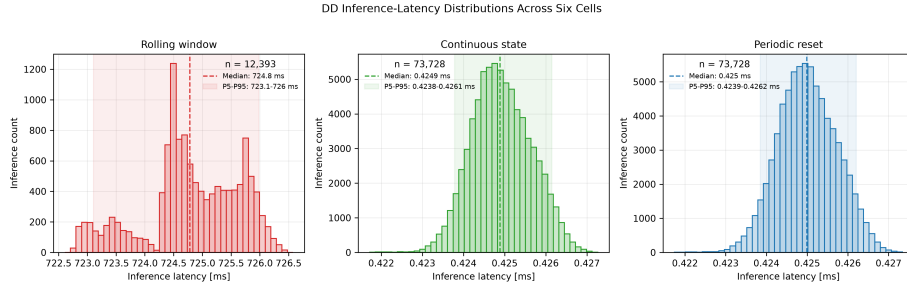


Figure A.24: DD latency distributions over the six hardware cell sequences. Exact rolling-window execution is shown separately from continuous-state and periodically reset execution because their scales and recurrent semantics differ substantially.

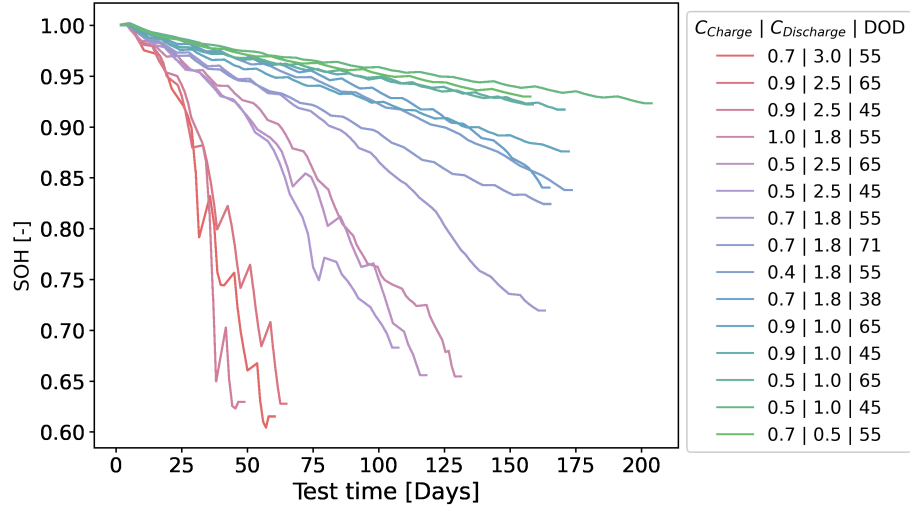


Figure A.25: Measured SOH trajectories across the complete LFP aging campaign. The campaign is divided into a six-cell training set, a three-cell validation set, and a six-cell independent holdout test set. The legend reports the applied charge C-rate, discharge C-rate, and depth of discharge for each trajectory. The plotted SOH values originate from repeated capacity checkups and are interpolated between checkup anchors. The exact cell-to-set assignment and measured coverage are reported in Table A.4.

Table A.4: Measured coverage and exact cell-to-set assignment for the fixed training, validation, and independent holdout test split. The 95th-percentile absolute C-rate is calculated over each complete 1-Hz trajectory. Load classes are defined only for the holdout test set. Its high-load class contains one cell and is exploratory.

Cell	Split	Duration [d]	SOH range	T range [°C]	P95 $ C_{\text{rate}} $	Test-set load
C01	Training	198.6	0.922–1.000	24.3–29.4	1.01	–
C03	Training	153.3	0.921–1.000	25.9–30.7	1.00	–
C05	Training	166.1	0.917–1.000	25.2–30.6	1.00	–
C11	Training	113.2	0.656–1.000	25.6–42.8	2.48	–
C17	Training	158.7	0.719–1.000	25.8–34.9	1.75	–
C23	Training	167.8	0.837–1.000	26.1–37.2	1.74	–
C07	Validation	167.6	0.874–1.000	25.2–32.0	1.01	–
C19	Validation	160.4	0.840–1.000	26.3–33.9	1.75	–
C21	Validation	160.8	0.824–1.000	26.2–37.6	1.75	–
C09	Test	102.6	0.682–1.000	25.0–41.1	2.51	Middle
C13	Test	44.7	0.623–1.000	24.9–42.0	2.50	Middle
C15	Test	60.7	0.627–1.000	25.1–43.6	2.53	Middle
C25	Test	127.1	0.655–1.000	25.8–37.1	1.76	Low
C27	Test	152.0	0.930–1.000	26.0–31.9	0.70	Low
C29	Test	56.0	0.604–1.000	25.5–49.4	3.00	High*

1066 Appendix B. HECM Lookup-Table Sensitivity

1067 The cross-model benchmark evaluates one fixed HECM implementation whose
1068 lookup surfaces were identified exclusively from the training and validation data
1069 and then held constant. This supplementary HECM-only analysis tests whether
1070 its disturbance response depends strongly on local lookup calibration changes.
1071 The R_i , R_1 , and R_2 surfaces are jointly scaled by -10% or $+10\%$. The τ_1 and
1072 τ_2 surfaces are likewise jointly scaled by -10% or $+10\%$, and the OCV surfaces
1073 are shifted by -10 mV or $+10$ mV. These six one-at-a-time perturbations and
1074 the nominal lookup are evaluated across 22 measurement and signal-integrity
1075 subcases plus the paired initialization intervention, all original seeds, and all 16
1076 frozen windows. The additive current-offset extension uses the same ± 0.05 A
1077 inputs as the cross-model benchmark. The combined design comprises 9184
1078 HECM executions.

1079 Main-campaign disturbances use their lookup-matched main baseline, the
1080 one-hour dropout uses a lookup-matched 48-h baseline, and initialization re-
1081 covery uses a separate paired lookup-matched baseline. For each disturbance,
1082 the lookup interaction is defined as the scenario Δ MAE under the perturbed
1083 lookup minus the corresponding Δ MAE under the nominal lookup. Results are
1084 first averaged within cells and then aggregated using the same equal-weight cell
1085 bootstrap as the primary benchmark. The analysis remains outside the cross-
1086 model robustness score because it probes one model’s local parameter sensitivity
1087 rather than a shared input disturbance.

1088 The local lookup changes shift baseline HECM MAE from 0.0314 to 0.0373,
1089 compared with 0.0337 for the nominal lookup. Across the 22 measurement and
1090 signal-integrity subcases, the largest absolute interaction is 0.006116 MAE. It
1091 occurs for the -0.05 A current offset under the $+10\%$ time-constant scaling,
1092 and its cell-bootstrap interval includes zero. The interaction is small relative to
1093 the nominal-lookup adverse current-offset penalty of 0.1763 MAE. The largest
1094 current-gain interaction remains 0.000443 MAE. Several smaller interactions
1095 have intervals that exclude zero, which identifies systematic local changes with-

out making the lookup calibration the dominant source of the observed disturbance penalties.

Initialization recovery exposes a separate boundary. Five lookup perturbations retain mean persistent recovery-or-censor times between 1.18 and 1.41 h, close to the nominal 1.20 h. Scaling all resistance surfaces by -10% instead increases the equal-weight cell mean to 5.10 h because one low-load holdout cell remains outside the recovery band through the 24-h horizon. The other five cells do not become censored. The lookup analysis therefore supports local stability of most measurement-disturbance responses, but it does not establish lookup independence of absolute accuracy or observer re-anchoring.

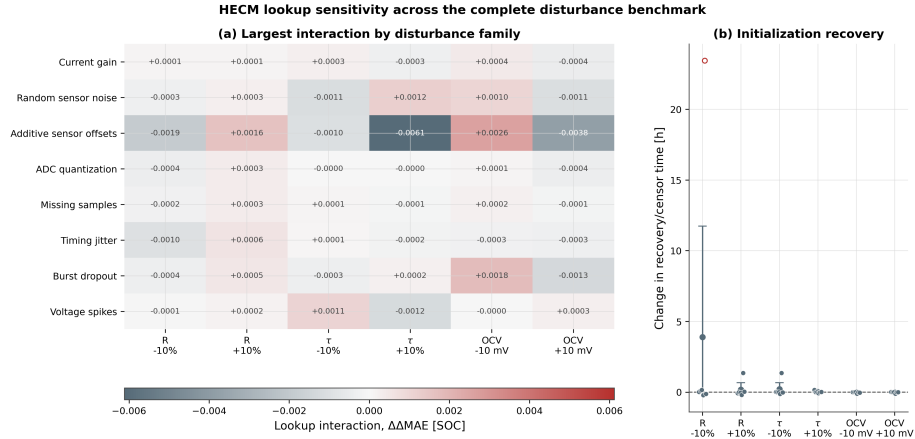


Figure B.26: HECM lookup sensitivity across the complete disturbance benchmark. Panel (a) reports the largest lookup interaction within each of the eight measurement and signal-integrity robustness families under local resistance, time-constant, and OCV perturbations. The additive-offset family includes current, voltage, and temperature offsets. Panel (b) reports the corresponding change in persistent initialization recovery-or-censor time. Points show equal-weight cell means. Open red points identify cell means containing a censored recovery window. Each run uses the appropriate lookup-matched main, dropout, or recovery baseline.

Table B.5: HECM lookup sensitivity across the complete disturbance benchmark. The largest robustness interaction is selected from 22 measurement and signal-integrity subcases after equal-weight cell aggregation. Bracketed counts report subcases whose 95% interaction interval excludes zero. Recovery is the persistent recovery-or-censor time from the paired initialization analysis, with the equal-weight censored-cell percentage in brackets.

Lookup condition	Baseline MAE	Max. robust $ \Delta\Delta\text{MAE} $ [CI]	Source subcase	Recovery/censor [h] [censored]
Nominal lookup	0.0337	Reference [-]	–	1.20 [0%]
Resistance -10%	0.0351	0.00191 [6/22]	Current offset +50 mA	5.10 [17%]
Resistance $+10\%$	0.0327	0.00162 [5/22]	Current offset +50 mA	1.38 [0%]
Time constants -10%	0.0343	0.00113 [6/22]	Current noise 0.10 A	1.41 [0%]
Time constants $+10\%$	0.0335	0.00612 [5/22]	Current offset –50 mA	1.22 [0%]
OCV -10 mV	0.0314	0.00260 [6/22]	Current offset +50 mA	1.18 [0%]
OCV $+10$ mV	0.0373	0.00378 [5/22]	Current offset +50 mA	1.19 [0%]

Appendix C. Supplementary Numerical Results

Table C.6: Six-cell baseline cell-macro statistics with hierarchical 95% confidence intervals.

Estimator	MAE [SOC]	RMSE [SOC]	P95 error [SOC]
DM	0.0690 [0.0499, 0.0823]	0.0740 [0.0532, 0.0881]	0.1088 [0.0768, 0.1307]
HDM	0.0412 [0.0301, 0.0515]	0.0442 [0.0319, 0.0557]	0.0649 [0.0449, 0.0846]
HECM	0.0337 [0.0246, 0.0434]	0.0388 [0.0286, 0.0493]	0.0644 [0.0468, 0.0821]
DD	0.0258 [0.0189, 0.0325]	0.0300 [0.0223, 0.0366]	0.0508 [0.0375, 0.0619]

Table C.7: Exact paired baseline-MAE comparisons across the six-cell holdout test set. Differences are defined as estimator B minus estimator A. Holm correction is applied across the six model-pair tests.

Estimator A	Estimator B	Mean difference [SOC]	95% CI	d_z	$p_{\text{exact}} / p_{\text{Holm}}$
DM	HDM	-0.0278 [-0.0381, -0.0171]	-1.93	0.0313 / 0.1875	
DM	HECM	-0.0352 [-0.0470, -0.0224]	-2.00	0.0313 / 0.1875	
DM	DD	-0.0431 [-0.0558, -0.0286]	-2.29	0.0313 / 0.1875	
HDM	HECM	-0.0075 [-0.0162, 0.0015]	-0.61	0.1563 / 0.1875	
HDM	DD	-0.0154 [-0.0243, -0.0084]	-1.39	0.0313 / 0.1875	
HECM	DD	-0.0079 [-0.0176, -0.0005]	-0.68	0.0938 / 0.1875	

Table C.8: Selected cell-macro scenario penalties relative to paired baseline. Current-gain and additive current-offset values use adverse directions from their matched signed sweeps.

Scenario	DM	HDM	HECM	DD
Current noise, $\sigma_I = 0.10$ A	-0.0009	+0.0000	+0.0038	-0.0000
Current gain, adverse 3%	+0.0108	+0.0101	+0.0060	+0.0038
Current offset, adverse 50 mA	+0.2467	+0.2727	+0.1763	+0.0401
Voltage offset, +0.02 V	+0.0000	+0.0005	-0.0040	+0.0038
Temperature offset, +3 °C	+0.0000	+0.0002	+0.0002	+0.0074
ADC quantization	+0.0038	+0.0035	-0.0018	+0.0002
Random missing samples, 2%	+0.0080	+0.0081	+0.0039	+0.0026
Timing jitter, ± 0.9 s	-0.0003	+0.0003	+0.0006	+0.0003
Burst dropout, 1 h	+0.0050	+0.0209	+0.0239	-0.0093
Voltage spikes, ± 0.20 V	+0.0000	-0.0003	+0.0002	-0.0001

Table C.9: Sensitivity of the illustrative normalized robustness score to the aggregation rule.

Estimator	Family balanced	Equal scenario	Highest level plus offsets
DM	0.452	0.549	0.573
HDM	0.351	0.476	0.469
HECM	0.360	0.552	0.562
DD	0.967	0.793	0.693

Table C.10: Canonical paired initialization recovery and target-hardware summary. Recovery values are equal-weight cell means over the common evaluation interval. Endpoints already satisfied when all estimators first provide valid output are recorded as conservative upper bounds.

Metric	DM	HDM	HECM	DD
First 300-s entry/censor time [h]	22.28	17.21	1.12	1.29
Persistent recovery/censor time [h]	22.28	22.28	1.20	2.60
Recovery excess-error AUC [SOC h]	1.735	1.738	0.078	0.116
First-entry censored fraction	0.89	0.67	0.00	0.00
Persistent-recovery censored fraction	0.89	0.89	0.00	0.00
Relapse fraction after first hold	0.00	0.22	0.17	0.78
Median SOC-core latency [μ s]	0.171	0.165	23.9	424.9 ^a
Flash [KiB]	27.0	27.1	470.1	115.9 ^a
Peak runtime RAM [KiB]	2.4	2.4	2.5	4.1 ^a

^aContinuous-state DD. Exact rolling-window execution requires approximately 724 ms and 67.3 KiB peak RAM.

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Data availability

The dataset is publicly available through its cited TU Berlin repository record [28]. Code, model configurations, benchmark definitions, random seeds, result summaries, hardware timing records, and analysis procedures are maintained in a public GitHub repository. The repository contains the neural model assets and HECM parameterization used for the reported experiments.

References

- [1] P. Shrivastava, P. A. Naidu, S. Sharma, B. K. Panigrahi, A. Garg, Review on technological advancement of lithium-ion battery states estimation methods for electric vehicle applications, *Journal of Energy Storage* 64 (2023) 107159. doi:10.1016/j.est.2023.107159.
- [2] F. Guo, L. D. Couto, G. Mulder, K. Trad, G. Hu, O. Capron, K. Haghverdi, A systematic review of electrochemical model-based lithium-ion battery state estimation in battery management systems, *Journal of Energy Storage* 101 (2024) 113850. doi:10.1016/j.est.2024.113850.
- [3] J. Yao, J. Kowal, Towards a smarter battery management system: A critical review on deep learning-based state of charge estimation of lithium-ion batteries, *Energy and AI* 21 (2025) 100585. doi:10.1016/j.egyai.2025.100585.
- [4] X. Li, J. Jiang, C. Zhang, L. Y. Wang, L. Zheng, Robustness of SOC Estimation Algorithms for EV Lithium-Ion Batteries against Modeling Errors and Measurement Noise, *Mathematical Problems in Engineering* 2015 (2015) 1–14. doi:10.1155/2015/719490.

- 1157 [5] F. Naseri, C. Barbu, A. Jensen, P. Setoodeh, J. R. Baena, N. Hevele,
1158 E. Schaltz, Toward Reliable Wireless BMS: Robust Battery State Estima-
1159 tion Under Noise and Uncertainty, IEEE Transactions on Vehicular Tech-
1160 nology (2025) 1–12doi:10.1109/TVT.2025.3638613.
- 1161 [6] F. Berger, D. Joest, E. Barbers, K. Quade, Z. Wu, D. U. Sauer, P. Dechent,
1162 Benchmarking battery management system algorithms - Requirements,
1163 scenarios and validation for automotive applications, eTransportation 22
1164 (2024) 100355. doi:10.1016/j.etrans.2024.100355.
- 1165 [7] Z. A. Khan, P. Shrivastava, S. M. Amrr, S. Mekhilef, A. A. Algethami,
1166 M. Seyedmahmoudian, A. Stojcevski, A Comparative Study on Different
1167 Online State of Charge Estimation Algorithms for Lithium-Ion Batteries,
1168 Sustainability 14 (12) (2022) 7412. doi:10.3390/su14127412.
- 1169 [8] Y. Gao, G. L. Plett, G. Fan, X. Zhang, Enhanced state-of-charge estima-
1170 tion of LiFePO4 batteries using an augmented physics-based model, Jour-
1171 nal of Power Sources 544 (2022) 231889. doi:10.1016/j.jpowsour.2022.
1172 231889.
- 1173 [9] Y. Gao, K. Liu, C. Zhu, X. Zhang, D. Zhang, Co-Estimation of State-of-
1174 Charge and State-of- Health for Lithium-Ion Batteries Using an Enhanced
1175 Electrochemical Model, IEEE Transactions on Industrial Electronics 69 (3)
1176 (2022) 2684–2696. doi:10.1109/TIE.2021.3066946.
- 1177 [10] J. Shen, W. Ma, J. Xiong, X. Shu, Y. Zhang, Z. Chen, Y. Liu, Alterna-
1178 tive combined co-estimation of state of charge and capacity for lithium-
1179 ion batteries in wide temperature scope, Energy 244 (2022) 123236. doi:
1180 10.1016/j.energy.2022.123236.
- 1181 [11] C. Wang, N. Cui, Z. Cui, H. Yuan, C. Zhang, Fusion estimation of lithium-
1182 ion battery state of charge and state of health considering the effect of
1183 temperature, Journal of Energy Storage 53 (2022) 105075. doi:10.1016/
1184 j.est.2022.105075.

- 1185 [12] P. Mondal, D. Bhavsar, K. Mittal, M. Mittal, Estimating State-of-Charge in
1186 Lithium-Ion Batteries Through Deep Learning Techniques: A Comparative
1187 Evaluation, *IEEE Access* 12 (2024) 78773–78786. doi:10.1109/ACCESS.
1188 2024.3408220.
- 1189 [13] D. P. Pau, A. Aniballi, Tiny Machine Learning Battery State-of-Charge
1190 Estimation Hardware Accelerated, *Applied Sciences* 14 (14) (2024) 6240.
1191 doi:10.3390/app14146240.
- 1192 [14] S. Giazitzis, M. Sakwa, S. Leva, E. Ogliari, S. Badha, F. Rosetti, A
1193 Case Study of a Tiny Machine Learning Application for Battery State-
1194 of-Charge Estimation, *Electronics* 13 (10) (2024) 1964. doi:10.3390/
1195 electronics13101964.
- 1196 [15] Y. Mazzi, H. Ben Sassi, A. Gaga, F. Errahimi, State of charge estima-
1197 tion of an electric vehicle’s battery using tiny neural network embedded
1198 on small microcontroller units, *International Journal of Energy Research*
1199 46 (6) (2022) 8102–8119. doi:10.1002/er.7713.
- 1200 [16] M. A. Kamali, A. C. Caliwag, W. Lim, Novel SOH Estimation of Lithium-
1201 Ion Batteries for Real-Time Embedded Applications, *IEEE Embedded Sys-
1202 tems Letters* 13 (4) (2021) 206–209. doi:10.1109/LES.2021.3078443.
- 1203 [17] B. Gou, Y. Xu, X. Feng, An Ensemble Learning-Based Data-Driven
1204 Method for Online State-of-Health Estimation of Lithium-Ion Batteries,
1205 *IEEE Transactions on Transportation Electrification* 7 (2) (2021) 422–436.
1206 doi:10.1109/TTE.2020.3029295.
- 1207 [18] W. Li, N. Sengupta, P. Dechent, D. Howey, A. Annaswamy, D. U. Sauer,
1208 Online capacity estimation of lithium-ion batteries with deep long short-
1209 term memory networks, *Journal of Power Sources* 482 (2021) 228863. doi:
1210 10.1016/j.jpowsour.2020.228863.
- 1211 [19] F. Rzepka, P. Hematty, M. Schmitz, J. Kowal, Neural Network Architecture

- for Determining the Aging of Stationary Storage Systems in Smart Grids,
Energies 16 (17) (2023) 6103. doi:10.3390/en16176103.
- [20] BQ79616 16-Series Battery Monitor, Balancer, and Integrated Hardware
Protector (2023).
- [21] Design Considerations of Current Sensing With BQ769x2 Family for Bat-
tery Management System (2022).
- [22] TMCS1108 3% Functional Isolation Hall-Effect Current Sensor With ± 100 -
V Working Voltage (2024).
- [23] TLE4972 high precision coreless current sensor for automotive applications
(2023).
- [24] O. Aiello, Electromagnetic Susceptibility of Battery Management Systems’
ICs for Electric Vehicles: Experimental Study, Electronics 9 (3) (2020) 510.
doi:10.3390/electronics9030510.
- [25] M. Liu, D. Geißler, S. Bian, B. Zhou, P. Lukowicz, Assessing the Impact of
Sampling Irregularity in Time Series Data: Human Activity Recognition
As A Case Study (Jan. 2025). arXiv:2501.15330, doi:10.48550/arXiv.
2501.15330.
- [26] 10s battery pack monitoring, balancing, and comprehensive protection, 50-
A discharge reference design (2024).
- [27] A. Aqachtoul, K. Karam, A. Elamrani, Y. Alidrissi, M. Najib, A. Rochd,
M. Bakhouya, MGFARM: An IoT and Edge-Driven Framework for Smart
and Sustainable Agricultural Practices, in: Y. Mejdoub, A. Elamri, M. Kar-
douchi (Eds.), Connected Objects, Artificial Intelligence, Telecommunica-
tions and Electronics Engineering, Vol. 1584, Springer Nature Switzerland,
Cham, 2026, pp. 419–425. doi:10.1007/978-3-032-01536-5_64.
- [28] F. Rzepka, Design of Experiments for Cyclic Aging of LFP 18650 Cells: Im-
pact of Charge/Discharge C-Rates, Depth of Discharge, and Temperature
on Battery Degradation (Nov. 2025). doi:10.14279/DEPOSITONCE-24800.

- 1240 [29] K. Siebertz, D. Van Bebber, T. Hochkirchen, Statistische Versuchspla-
1241 nung, Springer Berlin Heidelberg, Berlin, Heidelberg, 2017. doi:10.1007/
1242 978-3-662-55743-3.